

# Who Gets Protection? Interprovincial Inequality in Social Protection after Land Expropriation in China

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## Abstract

When rural land is expropriated in China, compensation may combine cash with continuing social protection. Using China Household Income Project (CHIP) 2013 and 2018 data on 2,239 land-expropriated households in 15 provinces, we study how compensation form varied across provinces and expropriation cohorts. Before 2008, moving from the 25th to the 75th percentile of provincial fiscal pressure was associated with a 20 percentage-point lower probability of reporting any continuing protection and a 17–19 percentage-point higher probability of cash only. From 2008 onward, these estimated relationships with fiscal pressure are markedly weaker, although imprecisely estimated. Across twelve consistently observed provinces, gaps in compensation form narrowed by about 38 percent, while the share of provincial variation explained by fiscal pressure fell from roughly one-quarter to less than 3 percent. The transition coincides with stronger national funding and approval rules and the expansion of rural social insurance. Provincial differences remained, but their earlier ordering by fiscal pressure weakened markedly.

**Keywords:** land expropriation; social protection; interprovincial inequality; compensation; fiscal capacity; rural China; CHIP

**JEL codes:** H75; H77; I38; O15; Q15

# 1 Introduction

Land expropriation has affected tens of millions of rural residents in China (Cai 2016; Zhan 2026). Rural land is collectively owned, and households hold contracted use rights rather than private ownership (NPC Standing Committee 1998, 2002). These rights support agricultural production and provide a fallback in old age and other adverse circumstances (Yu et al. 2010). Expropriation therefore removes both a productive resource and a source of economic security. Compensation matters not only for the amount households receive, but also for its form. A cash payment provides liquidity and resources for self-insurance; pension and medical-insurance arrangements link compensation to future income and health risks. Their relative value depends on households’ liquidity needs and exposure to future risks.

This creates a distributional question that a comparison of cash payments alone cannot answer. National rules established compensation obligations, but provincial and local governments shaped the schemes through which those obligations were implemented. Local financing and administrative capacity could therefore influence whether households reported protection arrangements alongside their compensation or cash alone. Spatial differences have long been central to the study of inequality in China, including the distribution of household resources and the organization of social provision (Knight and Song 1993; Li and Sicular 2014; Huang 2015). We examine a specific margin of that inequality: was protection reported after land expropriation systematically less common in fiscally weaker provinces, where the province’s own general-budget revenue covers a smaller share of its budget spending, and did this disadvantage persist across expropriation cohorts?

Existing research establishes the importance of land loss and compensation for household outcomes but provides a different view of this distributional margin. Studies examine income, health, satisfaction, and political trust after expropriation (Qu et al. 2018; Wang et al. 2019; Zhao and Xie 2022; Huang et al. 2024). Cai (2016) examines how local governments choose compensation arrangements, while Zhan (2026) traces changes in nationally dominant compensation regimes. Guo and Zhao (2022) study regional differences in general pension enrollment. We instead compare the composition of compensation reported by households across provinces and expropriation cohorts. Our interest is both in how far provinces differed and in whether those differences followed an identifiable fiscal and economic hierarchy.

We use the rural samples of the China Household Income Project (CHIP) in 2013 and 2018. Harmonizing the land-expropriation modules yields 2,239 households in fifteen provinces. Both surveys record the year and compensation associated with the household’s most recent expropriation; the five-cohort comparison covers 2,122 households with episodes from 2000 to 2019. Our primary measure, protection breadth, counts pension arrangements, medical insurance, and applications for Dibao, China’s means-tested minimum living guarantee. A broader indicator, any continuing protection, additionally includes housing and job arrangements, while cash only identifies households reporting a cash payment and none of these items. We refer to these

collectively as reported protection arrangements: the Dibao item records an application, not benefit receipt. The measures describe participation and arrangements reported in connection with expropriation rather than the monetary value of benefits.

We characterize provinces by fiscal pressure in 2000–02, defined as the share of general-budget expenditure in the province not covered by the province’s own general-budget revenue. Higher fiscal pressure indicates weaker own-source fiscal capacity. Because this measure is strongly correlated with provincial income, we interpret it as a marker of fiscal and economic capacity rather than an isolated fiscal mechanism. We first estimate its relationship with reported compensation within each cohort and how that relationship changes across cohorts. We then measure the average absolute difference in adjusted outcomes between two provinces, using the twelve provinces observed with sufficient support in every cohort. The first comparison describes a systematic provincial disadvantage; the second measures overall interprovincial inequality.

Our main findings are as follows. First, among households expropriated in 2000–03 and 2004–07, reported protection was substantially less common in fiscally weaker provinces. Moving from the 25th to the 75th percentile of 2000–02 fiscal pressure is associated with a roughly 20 percentage-point lower probability of reporting any continuing protection, against cohort means of 37 to 43 percent, and a 17 to 19 percentage-point higher probability of cash-only compensation. Second, this early fiscal disadvantage is much weaker among households expropriated from 2008 onward: the later point estimates are small and of mixed sign, although their confidence intervals are wide. In the pooled comparison, the estimated changes offset approximately 19 percentage points of the early difference in reported protection and 18 percentage points of the difference in cash only. The relationship remains steep in the 2004–07 cohort, placing the main change in the late 2000s rather than throughout the preceding decade.

Third, provinces also became more similar overall. Average provincial gaps in protection breadth, any continuing protection, and cash-only compensation fall by about 38 percent on the balanced twelve-province panel. For the broader protection indicator, the average gap falls from about 30 to 19 percentage points. Yet the relationship with 2000–02 fiscal pressure weakens more markedly than overall differences do. Fiscal pressure accounts for 24 to 31 percent of the variation in provincial protection breadth among early cohorts, compared with less than 3 percent among later cohorts. Substantial provincial differences remain, while the systematic disadvantage associated with fiscal weakness is much smaller. This distinguishes the weakening of a systematic provincial disadvantage from a general reduction in provincial differences.

Several checks clarify the scope of these findings. The narrowing of provincial gaps persists after corrections for sampling noise and reweighting for household composition. Alternative treatment of Dibao applications and general pension or medical-insurance labels preserves the broad early relationship and the narrowing of gaps, although the magnitude and statistical precision of the pooled change are sensitive to the treatment of pensions labeled NRPS for expropriations before its national rollout, and removing all cooperative-medical-scheme labels

halves the 2004–07 difference. The cash evidence offers no indication that less reported protection was offset by larger total payments. Among early-cohort households reporting a positive amount, the same fiscal-pressure interquartile range is associated with about 31 percent less total real cash compensation; the difference per unit of land is smaller and statistically imprecise. An illustrative comparison using external pension benchmarks also shows that the value of a pension arrangement varies substantially by scheme. Reported arrangements and benefit adequacy are therefore distinct distributional outcomes.

The institutional setting offers an explanation for why fiscal weakness could be associated with less continuing protection, rather than a preference for future promises over cash. Pension arrangements can require contributions at the time of expropriation, and national guidance in 2006–07 required funding to be secured as part of the expropriation process (General Office of the State Council 2006; MLSS and MLR 2007). Promising future benefits need not be inexpensive today. The late-2000s change coincides with stronger national approval and funding requirements and, from 2009, the introduction of a centrally co-financed rural pension (State Council 2009). These arrangements provided plausible routes for weakening the connection between local fiscal resources and reported protection. Their timing supports an institutional interpretation of the change, while the cohort comparisons do not establish their causal contribution or separate them from concurrent economic changes.

The paper makes three contributions. First, it treats compensation composition as a distributional outcome in its own right, complementing work on the subsequent consequences of land loss. Second, it distinguishes the narrowing of provincial differences from the weakening of the fiscal hierarchy associated with those differences. This adds a margin to research on the distributional role of social protection in China, beyond participation rates and the redistributive effects of transfers (Li et al. 2020a; Guo and Zhao 2022; Yang and Zhao 2024). Third, it constructs comparable measures from two CHIP land-expropriation modules and documents their wording and coding, extending the use of harmonized CHIP data to a different aspect of distributional change (Luo et al. 2020; Li et al. 2020b; Yang et al. 2021). The policy implication is that compensation should be assessed along both its monetary and institutional dimensions. Monitoring only cash amounts misses differences in protection arrangements; monitoring only enrollment misses differences in benefit value. Household-level arrangements and the benefits they ultimately deliver are essential complements to formal scheme adoption in assessing the distribution of protection after land loss.

Section 2 describes the institutional setting and related literature. Section 3 presents the data and empirical approach. Sections 4 and 5 examine the fiscal-pressure relationship and overall provincial differences. Section 6 reports measurement checks and discusses interpretation. Section 7 concludes.

## 2 Institutional background and related literature

### 2.1 Rural land rights and statutory compensation

Under the Land Administration Law, land in rural areas and on the urban fringe belongs to farmers' collectives unless the law assigns it to the state, and this includes residential plots (NPC Standing Committee 1998, art. 8). Members of the collective contract the land for agricultural use. The 1998 law set the contract term at thirty years and protected the household's contracted rights, but it still allowed land to be readjusted between individual households during the term with the consent of two-thirds of the village assembly and approval by the township and county (NPC Standing Committee 1998, art. 14). The Rural Land Contracting Law of 2002 tightened this protection. It confirmed the thirty-year term for arable land, required governments at county level or above to issue contract certificates, barred the collective from repossessioning land during the term, and prohibited readjustment except in special circumstances such as serious natural damage (NPC Standing Committee 2002, arts. 20, 23, 26, 27). A household that moved to a city with urban districts and took non-agricultural household registration had to return its arable land (NPC Standing Committee 2002, art. 26).

These rights were economically meaningful but less secure than private ownership. In a mid-1990s survey of 215 villages in eight provinces, land had been reallocated 1.7 times per village on average since household farming began; 28 percent of villages had not reallocated since 1983, and 53.4 percent of cultivated land had been reallocated at least once (Brandt et al. 2002). Private plots were held on average twelve years longer than collectively controlled land, and the risk of reallocation reduced long-term soil investment (Jacoby et al. 2002). Other village evidence finds reallocations to be infrequent and mostly partial (Kung 2000). Contracted land was thus an insecure asset, but an asset nonetheless: it yielded income and a fallback that expropriation removes for the land taken.

Cash compensation followed a national formula based on the land's original use (NPC Standing Committee 1998, art. 47). For arable land, the land compensation fee was six to ten times the average annual output of the three preceding years, and the resettlement subsidy was four to six times that output for each agricultural person to be resettled, capped at fifteen times per hectare. Provinces could raise the resettlement subsidy when the two payments did not maintain the household's former standard of living, but their sum could not exceed thirty times annual output. Compensation for standing crops and attachments was set by provinces. The land compensation fee belongs to the village collective, while the resettlement subsidy is paid to whoever arranges resettlement; when no collective resettlement is needed it is paid to the individual or, with the individual's consent, used to pay that person's insurance premiums (State Council 1998, art. 26). The CHIP question on cash compensation lists exactly these components. Two consequences follow. The cash amount was tied to agricultural output rather than to the land's value after conversion, as Cai (2016) emphasizes. And no national rule fixed whether compen-

sation would also include continuing protection. That decision is where provincial differences could arise.<sup>1</sup>

## 2.2 Social protection for land-expropriated farmers

National rules on social protection for land-expropriated farmers became progressively stricter between 2004 and 2007, and a general national rural pension followed from 2009. The main steps were as follows.

- **October 2004.** The State Council required local governments to ensure that expropriation did not lower farmers' living standards. Farmers inside urban planning areas were to be brought into the urban employment system and covered by a social-security system. When statutory compensation had reached its cap and was still insufficient, local governments could subsidize it from revenue from the paid use of state land, referred to below as land revenue (State Council 2004).
- **April 2006.** Guidance issued by the State Council's General Office asked provinces to draft concrete measures. Farmers inside urban planning areas were to join urban social security, and those outside were to join existing rural pension, cooperative medical, and rural Dibao schemes where these existed. Protection was to be no lower than the local Dibao standard and set, among other principles, by what government finances could bear. The funds were to come from the raised resettlement subsidy and the land compensation fee, with any shortfall covered by local land revenue, and were to be allocated during the expropriation itself (General Office of the State Council 2006).
- **August 2006.** The State Council folded social-security costs into the cost of expropriation and stated that no expropriation should be approved when social-security funds were not in place (State Council 2006).
- **April 2007.** The labor and land ministries urged provinces without a scheme to issue one by the end of 2007. Funding was to be shared among the individual, the collective, and the local government, and the individual contribution could be deducted directly from the household's compensation. Funds had to be deposited in a dedicated account within three months of the approval of the compensation plan, and expropriations without a scheme or without funds in place were not to be forwarded for approval (MLSS and MLR 2007).
- **September 2009.** The New Rural Pension Scheme began as a pilot in 10 percent of counties, with nationwide coverage targeted for 2020. Its basic pension was 55 yuan a month, paid in full by the central government in central and western provinces and at 50 percent in eastern provinces. The guideline required the scheme to be coordinated with social protection for

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<sup>1</sup>The 1998 revision of the law took effect on 1 January 1999. The 117 households in our sample expropriated in 1995–99 enter only the reference group of the pooled comparison in Section 3.4.

land-expropriated farmers and folded accounts under the earlier contributory rural pension into NRPS accounts (State Council 2009).

- **2010–2011.** The Social Insurance Law was adopted in 2010 and took effect in July 2011 (NPC Standing Committee 2010). A pilot pension for urban residents without employment began in 2011 on the same central financing terms as NRPS (State Council 2011).
- **February 2014.** The State Council merged NRPS and the urban residents’ scheme into a single basic pension for urban and rural residents (State Council 2014).

Two features of this chronology matter for what follows. First, 2004–07 was not a period without national rules. National documents set obligations, funding sources, and approval conditions, while provinces decided scheme design, eligibility, benefit levels, and cost shares. The requirements tightened in steps and culminated in the approval condition and the end-2007 target of the April 2007 notice. Second, NRPS was a general rural pension, not a program for land-expropriated farmers. It matters here because it gave local governments a nationally co-financed scheme into which expropriated farmers could be enrolled.

### 2.3 Related literature and contribution

The paper is closest to three groups of studies. The first studies land expropriation in China. Microdata studies estimate its effects on income, labor allocation, consumption, health, satisfaction, and political trust (Qu et al. 2018; Wang et al. 2019; Zhao and Xie 2022; Huang et al. 2024). Wang et al. (2019) use the same CHIP 2013 items we use, but as regressors that explain household welfare; here the compensation bundle is the outcome to be explained, observed in two harmonized waves. Cai (2016) and Zhan (2026) study compensation form directly. Cai combines case studies with a 2008 survey and shows that local governments favored the benefits less costly to them, and Zhan uses policy texts and qualitative sources to date a national shift toward housing-based compensation to the late 2000s, the same period in which the provincial pattern we document changes. Our evidence comes from household reports in a national sample, and the objects differ: a change in the nationally dominant form of compensation does not imply a change in how compensation is distributed across provinces. Section 6.6 examines housing directly.

The second group studies social protection and regional inequality in China. Spatial differences have long been a central component of Chinese inequality (Knight and Song 1993; Li and Sicular 2014; Lovely et al. 2019), and the decentralization of social provision has produced large subnational differences in program design and implementation (Xu 2011; Huang 2015; Huang and Kim 2020). Studies of Chinese social security treat pensions and other transfers as distributional objects, measuring their contribution to income inequality and redistribution (Cai and Yue 2020; Li et al. 2020a; Yang and Zhao 2024), the inequality of old-age resources (Hanewald et al. 2021), and the effects of the design of minimum-income support (Song and

Cheng 2024). Evidence on the integration of urban and rural health insurance shows that national consolidation changed what rural households actually used (Huang and Wu 2020; Hao et al. 2025). Closest in question is Guo and Zhao (2022), who ask whether initial provincial pension enrollment continues to predict later enrollment. They find higher participation where local capacity is greater, including where the fiscal self-reliance ratio is higher, and a narrowing of regional differences after 2012. Their outcome is general enrollment, while ours is protection reported in connection with a specific asset loss, and the change we document is concentrated in households expropriated from 2008 onward, earlier than the narrowing they find.

The third group uses CHIP to describe distributional change and treats the construction of comparable measures across waves as research work in its own right (Luo et al. 2020; Li et al. 2020b; Yang et al. 2021; Wang and Jv 2023). We follow that approach for the land-expropriation modules. Work on program targeting and take-up shows more generally that the existence of a program does not imply receipt (Alatas et al. 2012; Finkelstein and Notowidigdo 2019), and Han and Gao (2019) study realized welfare receipt in rural China as a distributional outcome. Our outcome is on this household side: the arrangements households report when their land was taken, including an application in the case of Dibao, rather than what provincial rules provided on paper.

### 3 Data, measurement, and empirical approach

#### 3.1 CHIP sample and expropriation cohorts

The household data are the rural samples of CHIP 2013 and CHIP 2018 (China Institute for Income Distribution n.d.a, n.d.b), repeated cross-sections designed to study income distribution in China. Both waves contain a land-expropriation module that asks whether the household ever experienced an expropriation of collectively owned land, when it occurred, how much land was taken, and what compensation the household received. Households with more than one expropriation are instructed to answer for the latest one, so a household’s cohort is the timing of its most recently reported expropriation.

CHIP 2013 and CHIP 2018 were fielded in 2014 and 2019; their income questions refer to the preceding calendar year, while the land-expropriation module asks about the most recent expropriation the household has experienced, with no reference-year limit. The analysis sample keeps households that report an expropriation year no later than the interview year, whose most recent expropriation occurred in 1995 or later, and that answer all six compensation items: cash, pension, medical insurance, Dibao, housing, and a job arrangement. These rules retain 2,239 of the 2,484 households that report an expropriation, 1,244 in 2013 and 995 in 2018 (Appendix Table A.1); 59 of them were expropriated in the interview year itself. The cohort analysis groups the 2,122 households expropriated from 2000 onward into five cohorts: 2000–03, 2004–07, 2008–10, 2011–13, and 2014 onward. Of the 603 households in the 2014+ cohort, 42 are CHIP 2013

households expropriated in 2014. The expropriation involved arable land in 93.7 percent of cases; 1.2 percent involved only a residential plot and 5.1 percent land of other use (Section 6.6). The public CHIP files contain no survey weight that can be applied consistently to these modules, so all household statistics are unweighted.

For the dispersion analysis we use a balanced panel of the 12 provinces with at least five households in every cohort: Anhui, Chongqing, Gansu, Guangdong, Henan, Hubei, Hunan, Jiangsu, Shandong, Shanxi, Sichuan, and Yunnan. It contains 1,953 households with complete information on the head. Beijing, Inner Mongolia, and Liaoning have households in every cohort but fewer than five in at least one, and are excluded so that changes in province coverage do not change measured dispersion mechanically.

### 3.2 Measuring compensation composition

The module records six forms of compensation associated with the expropriation: cash compensation; a pension; medical insurance; an application for the minimum living guarantee (Dibao), China’s means-tested social-assistance program (Song and Cheng 2024); housing arranged by the government; and a job arranged by the government. The Dibao item asks whether the household applied for Dibao at the time of the expropriation, not whether it later received Dibao benefits. Section 6.2 compares application with receipt at the survey date.

We summarize these items with three main outcomes.

- **Protection breadth** counts pension, medical insurance, and Dibao application, the three protection items observed in both waves (0 to 3). It is the primary outcome.
- **Any continuing protection** equals one if the household reports a pension, medical insurance, Dibao application, housing, or a job arrangement.
- **Cash only** equals one if the household received cash compensation and reports none of pension, medical insurance, Dibao application, housing, or a job arrangement. It is the complement of any continuing protection among households that received cash.

Two further measures serve as supporting evidence: an indicator for **pension or medical insurance**, and a **summary index**. Following Kling et al. (2007), the index averages the standardized values of pension or medical insurance, cash with pension or medical insurance, pension, medical insurance, protection breadth, and minus cash only, each standardized to mean zero and unit standard deviation in the full analysis sample. The average is not re-standardized, so the index is measured in index units, the average of the standardized components. Appendix Table A.2 states every outcome as a formula over the questionnaire items.

Table 1 reports cohort means. The share of households with any continuing protection rises from 37 percent in 2000–03 to 57 percent in 2011–13 and is 49 percent in 2014+, and the share with cash only moves in the opposite direction. The unweighted sample average is therefore not monotone over time. Our object is the distribution of protection across provinces rather than a

**Table 1:** Sample and compensation composition by expropriation cohort

| Cohort    | <i>N</i> | Prov. | Cash only | Any continuing | Pension | Pension or medical | Breadth |
|-----------|----------|-------|-----------|----------------|---------|--------------------|---------|
| 2000–03   | 185      | 15    | 0.573     | 0.373          | 0.249   | 0.297              | 0.595   |
| 2004–07   | 330      | 15    | 0.509     | 0.433          | 0.203   | 0.376              | 0.582   |
| 2008–10   | 432      | 15    | 0.535     | 0.444          | 0.273   | 0.370              | 0.620   |
| 2011–13   | 572      | 15    | 0.407     | 0.568          | 0.414   | 0.498              | 0.899   |
| 2014+     | 603      | 15    | 0.449     | 0.488          | 0.352   | 0.391              | 0.731   |
| All 2000+ | 2,122    | 15    | 0.475     | 0.482          | 0.320   | 0.405              | 0.719   |

*Notes.* Land-expropriated rural households, CHIP 2013 and 2018, expropriation years 2000 and later. Entries are unweighted means. “Any continuing” is any continuing protection; “Breadth” is protection breadth, the number of pension, medical insurance, and Dibao application reported (0 to 3). Cohorts refer to the household’s most recently reported expropriation. The 2014+ cohort covers 2014–19 and includes 42 CHIP 2013 households expropriated in 2014.

claim of steady national expansion.

### 3.3 Cross-wave harmonization and measurement

A valid cohort comparison requires the two waves to measure the same thing, and the 2013 and 2018 questionnaires differ in two ways. First, item numbering shifts because CHIP 2018 inserts an unemployment-insurance question with no 2013 counterpart; we exclude that item from every outcome. Second, the two insurance-type items have different response options. For pension type (K08\_2), in 2013 code 2 combines urban employee and urban resident pensions, while in 2018 code 2 is the resident pension alone; in both waves code 1 is a pension set up for land-expropriated farmers and code 3 is NRPS. The medical-type item (K09\_2) differs in the same way. Neither type item enters the main outcome definitions. They are used in the measurement stress tests of Section 6.3; pension type also identifies the groups in the subtype descriptions and in the illustrative pension-value comparison of Section 6.5. In addition, neither wave’s data file carries variable labels, so the mapping from data columns to questions has to be established from the questionnaires themselves. Appendix Table A.3 reproduces the Chinese question wording as administered, the official English translation, and the harmonized variable each item feeds. Appendix Table A.4 gives both waves’ response options and the coding rule for each item.

The wording also determines how the outcomes should be read. In the Chinese questionnaire as administered, each compensation item begins with 被征地时, “at the time of the expropriation”; the official English questionnaire translates the same items as “due to this land expropriation.” We therefore interpret the outcomes as protection reported in association with the expropriation episode. The wording does not by itself show that the expropriation caused the protection, and Section 6.3 examines whether general program enrollment could be recorded as expropriation compensation.

### 3.4 Fiscal pressure and empirical comparisons

We characterize provinces by a fixed measure of fiscal pressure averaged over 2000–02,

$$FP_p = \frac{1}{3} \sum_{s=2000}^{2002} \left( 1 - \frac{R_{ps}}{E_{ps}} \right), \quad (1)$$

where  $R_{ps}$  and  $E_{ps}$  are the general-budget revenue and general-budget expenditure of all levels of government within province  $p$  in year  $s$ , from the *China Statistical Yearbook* (National Bureau of Statistics of China 2001–2008); revenue includes tax and non-tax items. Higher values mean that a province depended more on transfers to finance its spending, although the measure is not an exact share of central transfers. Across the 15 provinces it ranges from 0.163 in Guangdong to 0.700 in Gansu, with an interquartile range (IQR) of 0.207, computed over the 15 provinces with equal weights. It is strongly correlated with log GDP per capita in 2000–02 ( $r = -0.887$ ; Appendix Figure A.1) and with trade openness before the 2008 global financial crisis ( $r = -0.854$ ). We therefore read it as a summary of provincial fiscal and economic capacity rather than as a separately identified fiscal mechanism, and we call provinces with high values fiscally weaker. The 2000–02 window overlaps the 2000–03 cohort; for the 2004–07 cohort and all later ones, it clearly predates the expropriation.

The first comparison asks how strongly compensation differed between provinces with high and low fiscal pressure within each cohort  $c$ :

$$Y_{ipt} = \alpha_c + \beta_c FP_p + \lambda_t + \delta_w + \varepsilon_{ipt}, \quad (2)$$

where  $Y_{ipt}$  is an outcome for household  $i$  in province  $p$  expropriated in year  $t$ ,  $\lambda_t$  are expropriation-year effects, and  $\delta_w$  is a survey-wave effect when a cohort spans both waves. A negative  $\beta_c$  for protection breadth means that households in fiscally weaker provinces reported fewer protection items. We report  $\beta_c \times 0.207$ , the difference associated with moving from the 25th to the 75th percentile of fiscal pressure, in natural units: protection items for breadth and percentage points for the binary outcomes.

The second comparison asks how this relationship changed across cohorts. On all 2,239 households we estimate

$$Y_{ipt} = \alpha_p + \lambda_t + \delta_w + \theta_T (FP_p \times Transition_t) + \theta (FP_p \times Post_t) + \varepsilon_{ipt}, \quad (3)$$

where  $Transition_t$  marks expropriations in 2004–07 and  $Post_t$  marks expropriations from 2008 onward. Province effects  $\alpha_p$  absorb the level of fiscal pressure. The omitted group consists of households expropriated before 2004. The coefficient  $\theta$  is the change in the fiscal-pressure coefficient between households expropriated from 2008 onward and those expropriated before 2004, and  $\theta_T$  is the corresponding change for 2004–07. We estimate  $\theta_T$  rather than assigning the

2004–07 cohort to either period, because whether it resembles the earlier or the later cohorts is an empirical question.

Standard errors are clustered by province, with the usual small-sample adjustment (CR1). Conventional  $p$ -values and confidence intervals use the  $t$  distribution with  $G - 1$  degrees of freedom, where  $G$  is the number of provinces in the estimation sample. With only 15 clusters, conventional cluster-robust tests can over-reject, so our main  $p$ -values come from a restricted wild-cluster bootstrap- $t$  that imposes the null hypothesis and draws Rademacher weights by province, with 9,999 draws (Cameron et al. 2008; MacKinnon et al. 2023); every wild-bootstrap test in the paper uses the same draws. As a further check we use a province-label permutation check: we reassign the 15 fiscal-pressure values across provinces at random 9,999 times, re-estimate  $\theta$  each time, and report the share of draws in which the estimate is at least as large in absolute value as the actual one. Because fiscal pressure was not randomly assigned, this is a robustness check on the provincial pattern rather than a design-based test.

### 3.5 Cross-provincial dispersion and inference

The dispersion analysis asks how far apart provinces were, regardless of whether their differences were related to fiscal pressure. We first remove differences in household composition: each outcome is regressed on household-head age and its square, sex, education (an indicator for each of the nine attainment categories in the questionnaire), and survey wave, and the residuals are re-centered at the sample mean. The head is the person identified as head at the survey, and age is measured in the survey’s reference year. Two households with missing information on the head’s education are left out of this adjustment. We then average the adjusted outcome within each province and cohort, using cells with at least five households. Our main statistic is the average absolute gap between two provinces,

$$G_c = \frac{2}{P(P-1)} \sum_{p < q} |\bar{Y}_{pc} - \bar{Y}_{qc}|, \quad (4)$$

where  $\bar{Y}_{pc}$  is the adjusted mean of province  $p$  in cohort  $c$  and  $P = 12$ .  $G_c$  is the expected absolute difference between two provinces drawn at random from the panel, and it is the Gini mean difference of the provincial means (Yitzhaki and Schechtman 2013). It has the units of the outcome: protection items for breadth and percentage points for the binary outcomes. Provinces receive equal weight because the object is inequality across provinces rather than across households. We summarize the change as the percentage decline from the average of the two early cohorts to the average of the three later ones, and we also report the cross-provincial standard deviation.

Early province-cohort cells are smaller, and sampling noise in a provincial mean inflates measured dispersion. We therefore also report a noise-corrected standard deviation, which subtracts the average squared standard error of the provincial means from their variance before taking the

square root (Chetty and Hendren 2018). Each standard error is the within-cell standard deviation divided by the square root of the cell size, which treats households in a cell as independent; a negative difference would be set to zero, which does not occur. The appendix reports three further checks: the standard deviation of province intercepts in a household-level random-intercept model estimated within each cohort, a cell-size matching exercise that repeatedly shrinks the cells of every cohort after 2000–03, including 2004–07, to the size of the province’s 2000–03 cell, and bootstrap intervals. Finally, to connect the two parts of the analysis, we report the share of the cross-provincial variance in adjusted means that fiscal pressure accounts for in each cohort, the  $R^2$  of a regression of provincial means on fiscal pressure.

## 4 Provincial fiscal capacity and compensation

### 4.1 Fiscal capacity and early compensation

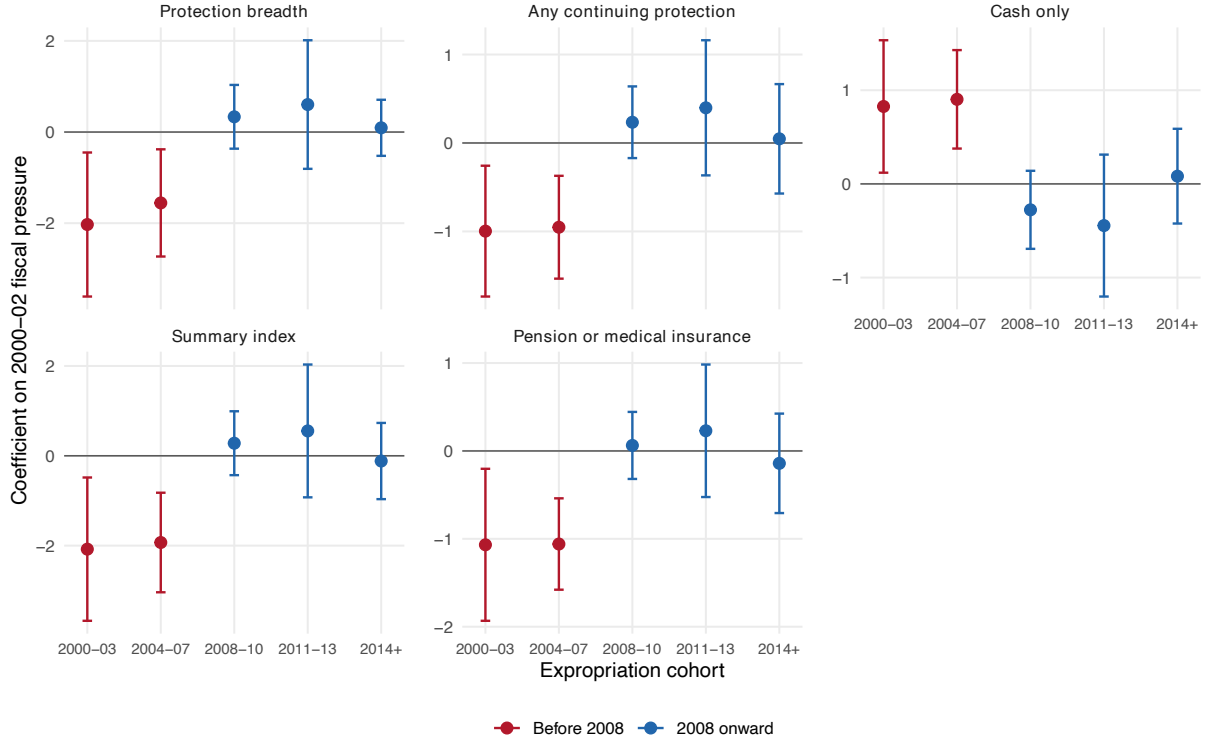
Figure 1 plots the coefficient  $\beta_c$  from equation (2) for each cohort and outcome. In the two early cohorts every outcome shows a large difference between fiscally weaker and fiscally stronger provinces. For households expropriated in 2000–03 and 2004–07, moving from the 25th to the 75th percentile of fiscal pressure is associated with 20.7 and 19.7 percentage points lower probability of any continuing protection, against cohort means of 37 and 43 percent, and with 17.1 and 18.7 percentage points higher probability of cash only. For protection breadth the corresponding differences are 0.42 and 0.32 items, against cohort means of about 0.6. The difference in any continuing protection is about half the cohort mean. Appendix Table A.5 reports the estimates.

The comparison does not rely on the cohort whose expropriation years overlap the measurement window of fiscal pressure. In 2004–07, when fiscal pressure clearly predates the expropriation, the coefficients are similar in size to those for 2000–03 and more precisely estimated. The same ordering appears when provinces are characterized by GDP per capita instead of fiscal pressure (Section 5.3).

### 4.2 Change after 2008

Among households expropriated from 2008 onward the estimated relationship is much weaker (Figure 1). For any continuing protection the interquartile differences in the three later cohorts are +4.9, +8.2, and +1.0 percentage points; for cash only they are −5.7, −9.2, and +1.7. None is statistically distinguishable from zero, and the signs do not form a consistent pattern. The estimates are not precise enough to show that the later relationship is zero: for any continuing protection the 95 percent intervals run from −4 to +13, −8 to +24, and −12 to +14 percentage points.

Table 2 reports the pooled comparison. In Panel A,  $\hat{\theta}$  is 1.625 for protection breadth, 0.937 for any continuing protection, and −0.847 for cash only, with wild-cluster bootstrap  $p$ -values of



**Figure 1:** Coefficient on 2000–02 provincial fiscal pressure, by expropriation cohort. Each point is  $\hat{\beta}_c$  from equation (2), estimated separately for each cohort with expropriation-year effects and, where a cohort spans both waves, a survey-wave effect. Bars are 95 percent confidence intervals based on province-clustered standard errors and the  $t$  distribution with  $G - 1$  degrees of freedom. Higher fiscal pressure means weaker own-source fiscal capacity, so a negative coefficient means that households in fiscally weaker provinces reported less of the outcome. Multiplying a coefficient by 0.207, the interquartile range of fiscal pressure across the 15 provinces, gives the difference between provinces at the 25th and 75th percentiles.

0.0053, 0.0252, and 0.0225. In interquartile terms, the change offsets 0.34 of the early 0.32–0.42 item difference in breadth, 19.4 percentage points of the early difference in reported protection, and 17.6 percentage points of the difference in cash only. These point estimates leave little of the early relationship, but they are imprecise: the 95 percent interval for the change in reported protection runs from 4.1 to 34.7 percentage points. The summary index changes by 1.472 ( $p = 0.0195$ ). The change for pension or medical insurance alone has the same sign but is less precisely estimated ( $p = 0.1943$ ). Province-label permutation  $p$ -values are larger, 0.0900, 0.0660, and 0.0582 for the three main outcomes. Panel B shows that the 2004–07 cohort resembles the earlier cohorts. The transition coefficients  $\hat{\theta}_T$  are  $-0.052$ ,  $-0.227$ , and  $0.261$ , with wild-cluster  $p$ -values of 0.8981, 0.4297, and 0.1507. The relationship facing households expropriated in 2004–07 is statistically indistinguishable from that facing households expropriated before 2004, even though national requirements tightened during those years.

**Table 2:** Change in the fiscal-pressure coefficient across expropriation cohorts

| Outcome                                                                          | Coefficient | Cluster SE | Cluster $p$ | Wild $p$ | Permutation $p$ | IQR difference |
|----------------------------------------------------------------------------------|-------------|------------|-------------|----------|-----------------|----------------|
| <i>Panel A. Cohorts expropriated from 2008 onward, <math>\hat{\theta}</math></i> |             |            |             |          |                 |                |
| Protection breadth                                                               | 1.625       | 0.656      | 0.027       | 0.0053   | 0.0900          | +0.34 items    |
| Any continuing protection                                                        | 0.937       | 0.344      | 0.017       | 0.0252   | 0.0660          | +19.4 pp       |
| Cash only                                                                        | -0.847      | 0.282      | 0.009       | 0.0225   | 0.0582          | -17.6 pp       |
| Summary index                                                                    | 1.472       | 0.670      | 0.045       | 0.0195   | 0.1505          | +0.31 units    |
| Pension or medical insurance                                                     | 0.653       | 0.398      | 0.123       | 0.1943   | 0.2604          | +13.5 pp       |
| <i>Panel B. Cohorts expropriated in 2004–07, <math>\hat{\theta}_T</math></i>     |             |            |             |          |                 |                |
| Protection breadth                                                               | -0.052      | 0.357      | 0.886       | 0.8981   | –               | -0.01 items    |
| Any continuing protection                                                        | -0.227      | 0.278      | 0.428       | 0.4297   | –               | -4.7 pp        |
| Cash only                                                                        | 0.261       | 0.198      | 0.209       | 0.1507   | –               | +5.4 pp        |

*Notes.* Estimates of equation (3) on all 2,239 households with province, expropriation-year, and survey-wave effects. Panel A reports  $\hat{\theta}$ , the coefficient on fiscal pressure  $\times$  expropriated from 2008 onward; Panel B reports  $\hat{\theta}_T$ , the coefficient on fiscal pressure  $\times$  expropriated in 2004–07. Both are changes relative to households expropriated before 2004, including the 117 households expropriated in 1995–99. Standard errors are clustered by province (CR1, 15 clusters); cluster  $p$ -values use the  $t$  distribution with 14 degrees of freedom. Wild  $p$ : restricted wild-cluster bootstrap- $t$  with Rademacher weights, 9,999 draws, the same draws for both panels. Permutation  $p$ : province-label permutation check, 9,999 draws. Both lie on a 1/10,000 grid and are reported to four decimals. “IQR difference” multiplies the coefficient by 0.207, the interquartile range of fiscal pressure across the 15 provinces: protection items for breadth, percentage points for binary outcomes, and index units for the summary index.

### 4.3 Timing of the change

The data place the change at the end of the 2000s rather than at one exact year. Appendix Table A.6 and Appendix Figure A.2 re-estimate the change in the fiscal-pressure coefficient with the boundary between earlier and later cohorts placed in each year from 2006 to 2011. Boundaries in 2006, 2007, and 2008 all yield a change with a wild-cluster  $p$ -value below 0.05 for each of the five outcomes; with a boundary in 2011 none does. For every outcome, the estimated change is largest in absolute value at the 2008 boundary and smaller at each of the other boundaries examined, as expected if the change is concentrated in the late 2000s: an earlier boundary assigns steep 2006–07 cohorts to the later group, and a later one assigns 2008 cohorts to the earlier group. We read this profile descriptively. It is not a formal test for a structural break, and its peak does not establish that the change occurred in a single year.<sup>2</sup>

A finer split of the early years gives the same picture. For the summary index the cohort coefficient is  $-2.078$  in 2000–03,  $-1.881$  in 2004–05, and  $-1.880$  in 2006–07, before it moves to  $+0.280$  in 2008–10 (Appendix Table A.7). The fiscal-pressure relationship remained as steep in 2006–07 as in 2000–03, although the April 2006 guidance and the August 2006 funding condition were already in force. The change instead coincides with the cohorts expropriated after the end-2007 target and the approval condition of April 2007. Because the 2008–10 cohort also includes expropriations after the 2009 pension launch and the boundary profile is descriptive, these

<sup>2</sup>The single-boundary change at 2008 (2.027 for the summary index) is larger than the pooled estimate in Table 2 (1.472). The two differ in sample and in the reference group: the single-boundary profile uses households expropriated from 2000 onward and pools both early cohorts in the reference group, whereas equation (3) uses all households from 1995 and estimates 2004–07 separately.

**Table 3:** Decline in cross-provincial dispersion, balanced panel of 12 provinces

| Outcome                      | Average gap | Standard deviation | Noise-corrected SD |
|------------------------------|-------------|--------------------|--------------------|
| Protection breadth           | 38.5        | 41.8               | 46.5               |
| Any continuing protection    | 38.3        | 37.4               | 39.1               |
| Cash only                    | 37.6        | 35.1               | 35.7               |
| Summary index                | 38.1        | 41.5               | 45.9               |
| Pension or medical insurance | 32.1        | 33.5               | 35.3               |

*Notes.* Entries are percentage declines from the average of the 2000–03 and 2004–07 cohorts to the average of the 2008–10, 2011–13, and 2014+ cohorts. “Average gap” is the average absolute gap between two provinces in demographically adjusted province-cohort means (equation 4). “Standard deviation” is the cross-provincial standard deviation of the same means. “Noise-corrected SD” subtracts the average squared standard error of the provincial means from their variance before taking the square root. Appendix Table A.9 reports random-effects and cell-size-matching versions.

comparisons do not date the approval rule and the pension separately (Section 6.7).

## 5 Cross-provincial dispersion

### 5.1 Average provincial gaps

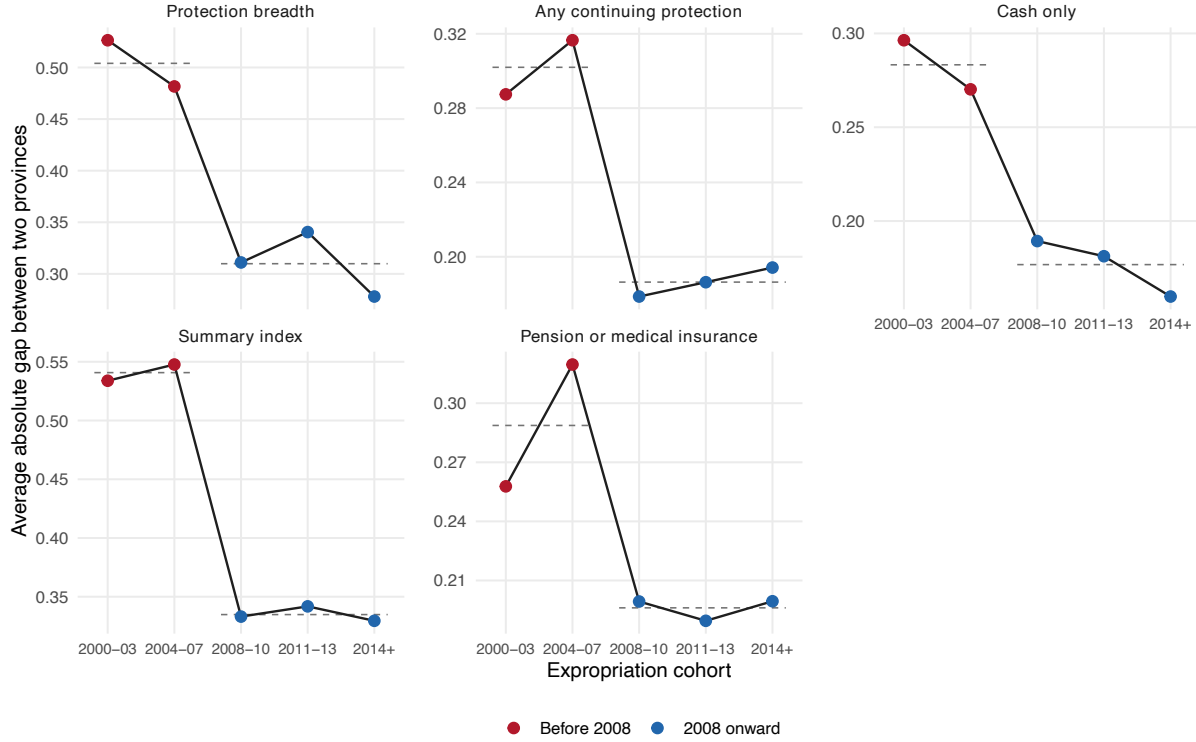
A smaller role for fiscal pressure does not by itself imply that provinces moved closer together. Provinces could cease to be ordered by fiscal capacity while remaining as far apart as before. This section therefore measures overall cross-provincial dispersion directly.

Figure 2 shows the average absolute gap between two provinces for each cohort. In protection breadth the gap is 0.526 and 0.482 items in the two early cohorts, about half a protection item, and 0.311, 0.340, and 0.278 in the three later cohorts, about one-third of an item. For any continuing protection it falls from 28.7 and 31.7 percentage points to 17.9, 18.6, and 19.4; for cash only from 29.6 and 27.0 to 18.9, 18.1, and 16.0. For all five outcomes each later cohort lies below both early cohorts (Appendix Table A.8). Table 3 summarizes the change: the average gap falls by 38.5 percent for protection breadth, 38.3 percent for any continuing protection, and 37.6 percent for cash only, and the standard deviation falls by similar proportions.

### 5.2 Sampling precision

Because early cells are smaller, sampling noise could make early provinces look further apart than they were. Adjusting for estimated noise does not reduce the decline. The noise-corrected standard deviation falls by 46.5 percent for protection breadth, 39.1 percent for any continuing protection, and 35.7 percent for cash only, compared with 41.8, 37.4, and 35.1 percent for the uncorrected standard deviation (Table 3). This adjustment rests on treating households within a province-cohort cell as independent.

Appendix Tables A.9 and A.10 report the remaining precision checks: a random-effects model, a cell-size matching stress test, bootstrap intervals, and leave-one-out estimates. Every

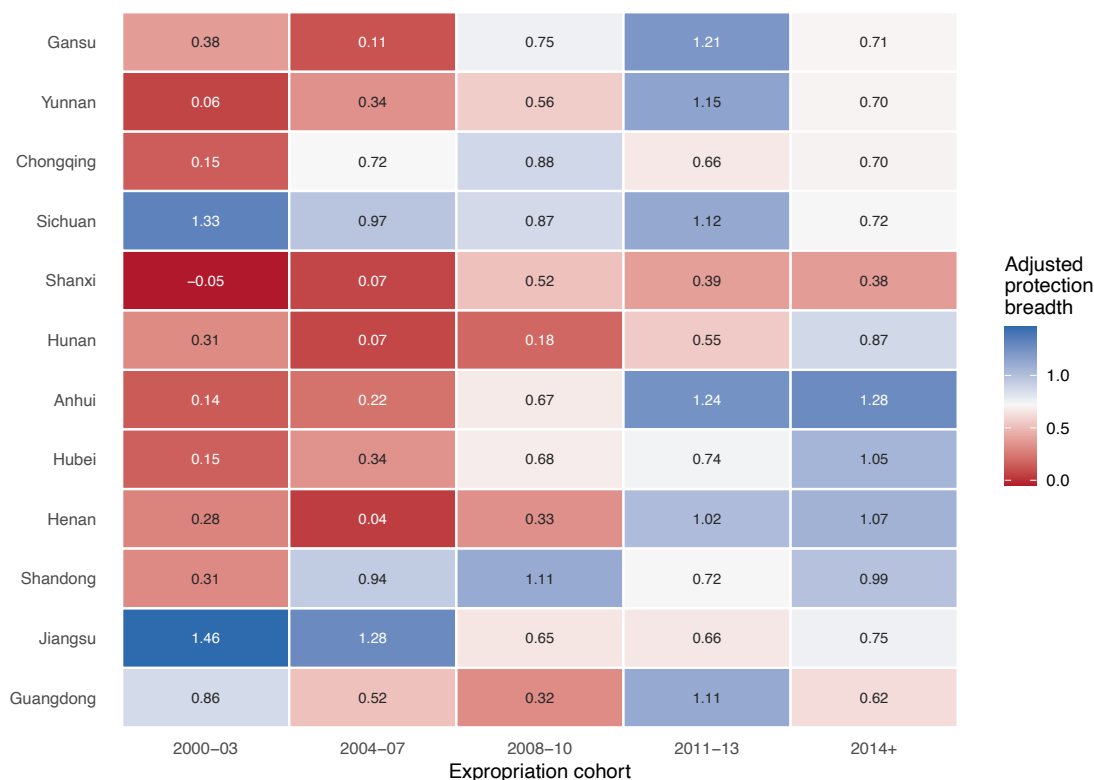


**Figure 2:** Average absolute gap between two provinces, by expropriation cohort, balanced panel of 12 provinces. Each point is  $G_c$  from equation (4), the average absolute difference between the demographically adjusted means of two provinces; it is measured in protection items for protection breadth, as a share for the binary outcomes, and in index units for the summary index. Dashed lines show the averages over the two earlier and the three later cohorts.

check shows a decline. When the 12 provinces themselves are resampled, the interval for protection breadth includes zero, so we state the dispersion result for the CHIP provinces rather than as a precise estimate for all of China.

### 5.3 Provincial ordering and remaining dispersion

The two measures move by different amounts. Overall dispersion falls by more than one-third, while the share of it that is ordered by fiscal capacity falls to a few percent. Figure 3 shows adjusted protection breadth by province and cohort, with provinces ordered by fiscal pressure. The early columns show a clear pattern, with low values concentrated among fiscally weaker provinces; later columns retain differences between provinces but no systematic ordering. In the two early cohorts, fiscal pressure accounts for 23.5 and 30.9 percent of the cross-provincial variance in adjusted protection breadth; in the three later cohorts it accounts for 0.1, 2.8, and 0.1 percent. Log GDP per capita gives the same profile, with  $R^2$  values of 0.240 and 0.327 early and 0.001, 0.061, and 0.000 later, and the summary index behaves the same way (Appendix Figure A.3). The differences not accounted for by fiscal pressure also narrowed, though by less:



**Figure 3:** Adjusted protection breadth by province and expropriation cohort, balanced panel of 12 provinces. Provinces are ordered by 2000–02 fiscal pressure, with the highest fiscal pressure (weakest own-source capacity) at the top. Cells report demographically adjusted province–cohort means; the color scale is centered on the full-sample mean.

the standard deviation of the residual from a regression of provincial means on fiscal pressure falls by 32.3 percent for protection breadth and 29.0 percent for cash only, against 41.8 and 35.1 percent for the total standard deviation.

What explains the provincial differences that remain after 2008? Part of the measured dispersion is sampling noise, for which the corrections in Section 5.2 adjust in both periods. The rest is not accounted for by fiscal pressure or GDP per capita (Appendix Figure A.3). Candidates include unmeasured provincial policies and local institutions, since eligibility and scheme details were set by provincial and local governments; the composition of expropriation projects, which differ in size, purpose, and type of land; differences in reporting; and sample selection. CHIP cannot separate these sources, because province–cohort cells are small and the survey records the county where the household lives rather than the county where the land was taken. What the data show is that after 2008 the remaining differences are at most weakly ordered by provincial fiscal capacity.

## 6 Robustness and interpretation

This section asks whether the two central results, the change in the fiscal-pressure relationship and the narrowing of provincial gaps, survive the measurement and sample checks that CHIP permits, and then discusses the institutional interpretation. Appendix Table A.13 collects the measurement stress tests.

### 6.1 Cross-wave measurement and questionnaire wording

Section 3.3 describes the harmonization. Two checks bear on the consistency of the later estimates across waves. First, the 2008–10 and 2011–13 cohorts are observed in both surveys with substantial support. Estimated separately by survey wave, the fiscal-pressure coefficients have the same sign in both waves and overlapping confidence intervals for all five outcomes in both cohorts (Appendix Table A.11), so the later pattern is not specific to one survey. Overlapping intervals do not show that the two waves measure identical objects, and the check does not cover the early cohorts. The 2014+ cohort also appears in both waves, but its CHIP 2013 part has 42 households, with five or more in only three provinces, and its wave-specific estimates are uninformative. Second, the insurance-type items, whose options differ between waves, enter none of the main outcome definitions; Appendix Table A.4 gives the mapping of their codes in each wave. They are used in the stress tests of Section 6.3, and pension type also identifies the groups in the subtype descriptions and the pension-value comparison. On wording, the Chinese items are anchored in time and the English translation attributes the protection to the expropriation (Section 3.3). The checks below ask whether the results depend on protections that may reflect general program enrollment.

### 6.2 Dibao

The Dibao item measures application at the time of the expropriation. Dibao application at expropriation is strongly associated with receipt at the survey date: households that applied are much more likely to have a member on Dibao when surveyed, 31.6 percent against 6.0 percent in CHIP 2013 and 20.0 percent against 4.0 percent in CHIP 2018 (Appendix Table A.12). But it is not receipt, and Dibao is means-tested social assistance rather than insurance. We therefore remove Dibao from every outcome (Appendix Table A.13). The early relationships are unchanged or slightly larger: the interquartile difference in any continuing protection is  $-22.3$  and  $-22.1$  percentage points in 2000–03 and 2004–07, and in cash only  $+17.8$  and  $+21.1$ . The decline in the average provincial gap remains 38.8, 35.4, and 31.5 percent for the three main outcomes. The pooled change  $\hat{\theta}$  keeps its sign and most of its size but is less precisely estimated, with wild-cluster  $p$ -values of 0.0276, 0.0492, and 0.0593.

### 6.3 Pension subtype reporting

The pension item asks whether a pension was arranged at the time of the expropriation; a follow-up question asks, in the present tense, what type the pension is. Of the 706 households reporting a pension, 58 percent describe it as NRPS and 24 percent as a scheme set up for land-expropriated farmers (Appendix Table A.14). Sixty-nine households report an NRPS label for an expropriation in or before 2008, before the 2009 national rollout of NRPS; we refer to them as pre-rollout NRPS-labeled reports. Three explanations are possible, and CHIP does not identify which applies to a given household. First, local schemes of the new type preceded the national program: the labor ministry began pilots of a new rural pension in eight counties in January 2006 (NDRC 2007), and Beijing introduced municipal new rural pension measures in 2007 before replacing them with an urban–rural residents’ pension in January 2009 (Beijing Municipal People’s Government 2008). Thirty-seven of the 69 reports concern expropriations in 2006–08. Second, the 2006 guidance directed farmers outside urban planning areas into the earlier contributory rural pension where it existed (General Office of the State Council 2006), and the 2009 guideline folded accounts under that pension into NRPS (State Council 2009), so a household enrolled in an older scheme at the time of the expropriation could now describe its pension as NRPS. Third, respondents may have misreported the type. The reports are not evenly spread: 47 of the 69 are in the four provinces with the lowest fiscal pressure, and 23 fall in the pre-2004 reference group of equation (3).

Our main estimates keep these households, because their answer to the pension question is internally consistent and only the type label is in doubt. Two stress tests show how much the results depend on them (Appendix Table A.13). The first addresses the concern that general NRPS enrollment was recorded as expropriation compensation: it counts no NRPS-labeled pension as a pension. This removes the protection most likely to reflect general enrollment, and it strengthens the main binary results. The pooled change is 1.061 for any continuing protection and  $-0.972$  for cash only, with wild-cluster  $p$ -values of 0.0060 and 0.0071, and the decline in the average provincial gap is 47.7, 40.0, and 38.5 percent for the three main outcomes. The second is an intentionally severe change of sample that drops all 69 households, including the 23 in the pre-2004 reference group, where they are concentrated among fiscally stronger provinces. It is not a corrected estimate. The early relationships and the narrowing of provincial gaps survive: the early interquartile differences are  $-17.4$  and  $-16.4$  percentage points for any continuing protection and  $+14.0$  and  $+16.2$  for cash only, and the average gap falls by 30.9, 32.5, and 32.3 percent. The pooled change becomes smaller and less precise, with wild-cluster  $p$ -values of 0.2435, 0.0768, and 0.0465, against 0.0053, 0.0252, and 0.0225 for the main estimates; combined with the removal of Dibao, none of the three pooled changes is significant. The early relationship and the narrowing of gaps are therefore more stable across these treatments than the precision of the pooled change, which depends on how the reports are treated. A final row of Appendix Table A.13 restricts the sample to the 2,180 households expropriated no later than the reference

year; the estimates are close to the main ones.

The medical item raises a parallel concern. The New Rural Cooperative Medical Scheme (NCMS) accounts for 42 to 92 percent of medical-insurance reports in the cohorts from 2000 onward, although it began only in 2003; 30 NCMS-labeled reports concern expropriations before 2003, so the same three explanations apply to them. A stress test that counts no NCMS-labeled report as medical insurance, 540 of the 676 medical reports, is deliberately severe, because most NCMS reports in the 2004–07 and later cohorts could reflect enrollment at the time of the expropriation. It is not a preferred coding. The pooled change and the narrowing of provincial gaps survive it:  $\hat{\theta}$  is 1.241, 0.974, and  $-0.828$  for the three main outcomes, with wild-cluster  $p$ -values of 0.0079, 0.0166, and 0.0201, and the average provincial gap falls by 42.7, 34.9, and 37.3 percent (Appendix Table A.13). The 2000–03 differences also survive, at  $-20.4$  percentage points for any continuing protection and  $+16.9$  for cash only. The 2004–07 differences roughly halve, to  $-11.2$  and  $+9.7$  percentage points, and are no longer statistically distinguishable from zero, so part of the early difference in that cohort runs through NCMS enrollment. Under this recoding, the pooled change, the 2000–03 differences, and the narrowing of gaps remain, while the 2004–07 differences do not.

#### 6.4 Household characteristics and sample composition

Household heads in later cohorts are older at expropriation. Reweighting the later cohorts to the early distribution of head’s sex, education categories, and age at expropriation and its square, using the inverse-probability weights of DiNardo et al. (1996), removes the main imbalance without extreme weights (the largest weight is 2.88). The decline in the average provincial gap remains 27 to 34 percent (Appendix Table A.15). The reweighting balances observed characteristics of the head identified at the survey and does not remove other differences in composition.

Recall periods are longer for early expropriations. We measure recall as the number of calendar years between the expropriation and the interview. No household in the 2000–03 cohort was expropriated within ten years of the interview, so this check uses 2004–07 as the early window. To keep provincial coverage constant, each restricted comparison is paired with an unrestricted comparison on the same provinces, those with at least five households in 2004–07 and in each later cohort under the restriction: 13 provinces for a ten-year limit and 8 for an eight-year limit. On these provinces, the decline in the average gap for the summary index is 27.8 percent without a restriction and 35.3 percent with the ten-year limit, and 12.9 and 33.9 percent for the eight-year comparison. The early fiscal-pressure coefficient remains large under the same restrictions (Appendix Table A.16; Appendix Figure A.4). The narrowing persists under shorter recall limits on these common-support subsets, although the eight-year comparison covers only eight provinces.

Three features of the sample limit what it represents. The questionnaire records only the most recent expropriation, so membership in an early cohort implies no later reported expro-

priation. CHIP records the survey province rather than the province of the expropriation. And households that left the rural sampling frame before the survey, for example after resettlement with urban household registration, are not observed. The estimates therefore describe rural-frame households at the survey date, located by their survey province.

## 6.5 Cash compensation value

If households in fiscally weaker provinces had received more cash in place of continuing protection, the early pattern would be a difference in form with little difference in generosity. The observed pattern provides no evidence that lower reported protection was offset by larger total cash payments. The module records the amount of compensation the household actually received (item K07\_3), which we deflate to 2018 yuan with the consumer price index of the expropriation year (World Bank 2026), assuming payment in that year, and the area taken. Between 84 and 94 percent of households in each cohort report an amount. Among households expropriated before 2008, moving from the 25th to the 75th percentile of fiscal pressure is associated with 0.37 log points (about 31 percent) less real compensation ( $p < 0.001$ ) and 0.11 log points less per mu of land taken (1 mu is one-fifteenth of a hectare, about 667 square meters), a difference that is not statistically significant (Appendix Table A.17). Among later cohorts both differences are positive and imprecise, and the pooled change is not significant (wild-cluster  $p$ -values of 0.2650 and 0.5596). Thirty-one households report receiving more than the total compensation for the land, which the questionnaire rules out; excluding them leaves the early difference at 0.37 log points. The value evidence thus shows no sign that higher total cash offset lower reported protection in the early cohorts. It observes total cash among households reporting a positive amount, not compensation for land of identical quality, and it is too imprecise to show how the value relationship changed later.

One feature of the rules makes this check conservative. Resettlement subsidies could pay insurance premiums, and individual pension contributions could be deducted directly from compensation (State Council 1998; MLSS and MLR 2007). Where households received continuing protection, the cash they report receiving may be net of such contributions. Since households in fiscally stronger provinces reported more continuing protection, deductions would lower their reported cash, which works against the finding that households in fiscally weaker provinces reported less cash.

CHIP does not observe household-specific pension benefits, but external benchmarks permit an illustrative comparison of magnitudes. In CHARLS 2013 (China Center for Economic Research and Institute of Social Science Survey 2015; Zhao et al. 2014), the median benefit among current NRPS recipients is 60 yuan a month. The median annual benefit is 648 yuan for NRPS and 5,398 yuan for pensions set up for land-expropriated farmers, both in 2010 prices. Discounted at a 2 percent real rate, with a ten-year wait, the first payment at the end of the eleventh year after the expropriation, and fifteen annual payments, these flows are worth 6,828

and 56,897 yuan (Appendix Table A.18). The median cash compensation of households reporting each pension type is 20,301 and 24,679 yuan in the same prices, so the NRPS stream is worth about 0.34 of the cash payment and the dedicated pension about 2.3 times it; with waits of 5 to 15 years and durations of 10 to 20 years at the same rate, the ratios range from 0.21 to 0.47 and from 1.46 to 3.24. Comparing a lump sum with the discounted value of an annuity stream is a standard benchmark (Mitchell et al. 1999; Büttler and Teppa 2007). Our figure is the discounted value of a benchmark pension flow under stated delay and duration assumptions for one beneficiary. It is not matched household pension wealth, net benefit value, or government cost; it omits mortality, indexation, and the insurance value of an annuity (Brown 2003); and it is not a welfare ranking of cash and pensions. What it shows is that the value of continuing protection depends heavily on its type. Most later pension reports carry the NRPS label, whose benchmark value is small relative to cash, which reinforces why a weaker link between fiscal capacity and reported protection does not imply equal protection value.

## 6.6 Alternative definitions, timing, and remaining checks

*Type of land.* Re-estimating on the 2,097 households whose expropriation involved arable land keeps all 12 provinces of the panel with at least five households in every cohort. The decline in the average gap is 38.6, 35.6, and 29.9 percent for the three main outcomes, and each of their pooled changes is larger than in the full sample, with wild-cluster  $p$ -values of 0.0138 or less (Appendix Table A.19). The share of arable-land cases does not differ significantly across cohorts at the 5 percent level ( $\chi^2 p = 0.114$ ) and is unrelated to fiscal pressure across provinces ( $r = 0.055$ ).

*Housing.* Zhan (2026) dates a shift toward housing-based compensation to the late 2000s. Housing cannot account for our results. Protection breadth never includes housing and shows the largest decline in dispersion among the three main outcomes. Only 17 of 2,122 households report housing as their only protection arrangement, and removing housing from any continuing protection leaves a decline of 33.5 percent in the average gap, against 38.3 percent with it. Housing receipt peaks at 7.6 percent in the 2004–07 cohort and falls to 2.0 percent in 2014+ (Appendix Table A.20). That last pattern may reflect a limitation of rural survey data rather than evidence against Zhan’s account, since households resettled into urban housing are the most likely to leave the rural sampling frame; CHIP does not observe those households.

*Alternative outcome definitions.* The decline in dispersion appears for each of the alternative definitions in Appendix Table A.21. It ranges from 24.8 percent for pension alone to 42.9 percent for medical insurance alone.

*The financial crisis and trade exposure.* The late 2000s also brought the global financial crisis, which hit provinces unevenly. Trade openness in 2005–07 is so highly correlated with fiscal pressure ( $r = -0.854$ ) and log GDP per capita ( $r = 0.872$ ) that the two cannot be separated across 15 provinces. Adding trade openness  $\times$  expropriated from 2008 onward to

equation (3), estimated on households expropriated from 2000 onward, reduces the estimated change in the fiscal-pressure coefficient by 41 to 72 percent (Appendix Table A.22). The pooled change is therefore sensitive to controlling for trade exposure, and the data cannot say which of the two correlated provincial characteristics it reflects. This affects the attribution of the change, not the dispersion result, which is computed directly from provincial means without either provincial characteristic.

*Provincial frameworks.* Provinces issued their own rules for land-expropriated farmers at different times. Beijing’s municipal rules on compensation and resettlement, which set pension and medical-insurance contributions for farmers converted to non-agricultural status, took effect in July 2004 (Beijing Municipal People’s Government 2004), and Jiangsu’s provincial measures on basic living security for land-expropriated farmers took effect in September 2005 (Jiangsu Provincial People’s Government 2005). We could not trace the first province-wide framework of every CHIP province to a dated primary document, and some provinces issued pilot or departmental rules before a province-wide one. Provincial adoption dates are therefore not used in the analysis.

## **6.7 Interpretation: fiscal capacity, funded approval, and national pension delivery**

Three institutional forces fit the evidence. They are related, but they operate through different channels.

*Early local fiscal capacity.* National law fixed cash compensation but left continuing protection to provinces and localities, and national guidance itself tied the level of protection to local finances. The April 2006 guidance asked local governments to set protection by, among other principles, what government finances could bear and to give full consideration to the capacity of local finances, village collectives, and farmers (General Office of the State Council 2006). Cai (2016) argues that richer localities also had more reason to go beyond the statutory cash formula. A province with little own-source revenue therefore had less money for up-front contributions, and national guidance allowed it to set protection at a level its finances could bear. In the early cohorts, households in fiscally weaker provinces reported less continuing protection and, among those reporting a positive amount, also less total cash. This pattern is consistent with an overall capacity constraint. Provincial comparisons cannot rule out substitution between forms of compensation within households.

The same institutional design helps explain why the observed pattern does not fit the simple prediction that fiscally constrained governments would prefer deferred pension promises to cash. A pension that pays only when the recipient reaches retirement age spreads the cost over time, and local officials describe this advantage (Cai 2016). Up-front contribution and funding requirements could make pension-linked compensation costly even when benefit payments occurred later. From 2006, national rules required social-protection funds to be secured dur-

ing the expropriation, drawn in part from the compensation fees, and in place before approval (General Office of the State Council 2006; State Council 2006); the April 2007 notice added that the funds be deposited in a dedicated account within three months of the approval of the compensation plan and that individual contributions could be deducted from the household's compensation (MLSS and MLR 2007). Before 2006, the 1998 regulations already allowed the resettlement subsidy to pay the household's insurance premiums (State Council 1998, art. 26), and Cai (2016) describes local schemes that tie pension eligibility to the amount of land taken. Such rules permitted, but did not guarantee, up-front financing, and formal requirements do not establish compliance. Continuing protection therefore need not have been a cheap deferred promise for a fiscally constrained government. The early cohorts show no sign of such deferral, including the 2000–03 cohort.

*Funded approval rules.* The April 2007 notice made the August 2006 funding condition operational in the approval process: provinces were urged to issue schemes by the end of 2007, the social-protection terms had to be disclosed to affected farmers before approval was sought, and expropriations without a scheme or without funds in place were not to be forwarded for approval (State Council 2006; MLSS and MLR 2007). Its timing is consistent with the data: the fiscal-pressure relationship was still steep for households expropriated in 2006–07 and much weaker for those expropriated in 2008–10. The 2008–10 cohort, however, includes expropriations both before and after the 2009 pension launch, so the grouped comparison cannot date the approval rule separately from the pension. A requirement enforced at approval raises the minimum provision in every province, and the requirement is most likely to bind in provinces that would otherwise have provided least.

*National pension delivery and co-financing.* From 2009 NRPS provided a nationwide pension into which land-expropriated farmers could be enrolled, and the 2009 guideline required coordination between the two. The central government paid the full basic pension in central and western provinces and half of it in eastern ones (State Council 2009), so the cost of a basic pension fell less on the provinces with the least own-source revenue. The pension-type reports are consistent with a growing role for this channel. NRPS labels account for 22 percent of pension reports in the 2000–03 cohort, all of them pre-rollout reports, and 65 to 68 percent in the 2011–13 and 2014+ cohorts, while pensions set up for land-expropriated farmers stay between 22 and 28 percent (Appendix Table A.14). The two later channels differ. The approval rule changed what local governments had to fund before an expropriation could proceed. NRPS changed who paid for the basic benefit and gave local governments a ready scheme to use. The cohort comparison does not separate the timing or the contributions of these two channels.

These were national policy changes, although local implementation and coverage differed across places and years; the province-cohort design therefore does not provide a credible untreated comparison group. The crisis-era expansion of credit and land finance coincides with them. We therefore interpret the evidence as institutional and descriptive rather than as a

causal estimate of any single rule.

## 7 Conclusion and discussion

This paper studies the form in which Chinese households were compensated for expropriated land and how it varied across provinces and expropriation cohorts, using harmonized land-expropriation modules from CHIP 2013 and 2018. The composition of compensation, cash-only compensation or bundles that include reported protection arrangements, is the outcome to be explained.

Three findings stand out. First, among households expropriated before 2008, reported protection was substantially less common, and cash-only compensation more common, in fiscally weaker provinces: across the interquartile range of 2000–02 fiscal pressure, the difference is about 20 percentage points for any continuing protection and 17 to 19 percentage points for cash only. Second, these estimated relationships with fiscal pressure are markedly weaker among households expropriated from 2008 onward, although the later estimates are imprecise. Third, provinces moved closer together overall, with the average absolute gap between two provinces falling by about 38 percent, but this decline is much smaller than the decline in the share of provincial differences associated with fiscal pressure. The observed cash amounts provide no evidence that lower reported protection in the early cohorts was offset by larger payments.

The timing points to national rules. The fiscal-pressure relationship remained steep through the 2006–07 cohort and was much weaker after the end-2007 target for funded provincial schemes; from 2009, a centrally co-financed rural pension added a national delivery channel. The cohort comparison cannot separate these arrangements from each other or from the economic changes of the late 2000s, but it shows that the earlier association between fiscal weakness and limited reported protection weakened markedly and that provincial gaps narrowed.

For policy, compensation should be assessed along both its monetary and institutional dimensions. National rules and financing arrangements can reduce the dependence of reported protection on local fiscal capacity, but a weaker link between fiscal capacity and reported protection does not by itself imply equal protection value. The basic rural pension paid 55 yuan a month at its launch (State Council 2009), and benchmark pension values differ widely by scheme (Appendix Table A.18). Monitoring should therefore follow the arrangements households obtain and the benefits they deliver, not only formal scheme adoption.

The evidence describes reported arrangements, not verified receipt, benefit value, or take-up, and covers households remaining in the rural sampling frame; Section 6 shows which conclusions are stable across measurement choices and which are sensitive to them. Administrative benefit records, complete expropriation histories, and eligibility information would sharpen these boundaries. Within CHIP’s rural sample, the strong fiscal ordering of compensation form in early cohorts was markedly weaker in later cohorts.

## Appendix A. Supplementary tables and figures

### A.1 Measurement

Table A.1 gives the construction of the analysis sample, and Tables A.2–A.4 define the outcomes and document the questionnaire items on which they are built.

**Table A.1:** Construction of the analysis sample

| Step                                                                  | Households | CHIP 2013 | CHIP 2018 | Excluded |
|-----------------------------------------------------------------------|------------|-----------|-----------|----------|
| Report an expropriation of collectively owned land (K05 = 1–4)        | 2,484      | 1,424     | 1,060     |          |
| Expropriation year reported and no later than the interview year      | 2,392      | 1,359     | 1,033     | 92       |
| Expropriation year 1995 or later                                      | 2,302      | 1,280     | 1,022     | 90       |
| All six compensation items answered (analysis sample)                 | 2,239      | 1,244     | 995       | 63       |
| of which: expropriated 2000 or later (cohort analysis)                | 2,122      | 1,154     | 968       |          |
| of which: head’s age, sex and education all observed                  | 2,237      | 1,244     | 993       |          |
| of which: 2000 or later with head’s characteristics observed          | 2,120      | 1,154     | 966       |          |
| of which: expropriated no later than the reference year (2013 / 2018) | 2,180      | 1,202     | 978       |          |

*Notes.* Rural households in CHIP 2013 and 2018 that report an expropriation of collectively owned land. CHIP 2013 was fielded in 2014 and CHIP 2018 in 2019, and the expropriation item has no reference-year limit. The year step keeps households reporting a calendar year no later than the interview year; of the 92 households it excludes, 60 report no year and 32 report a code that is not a year, and none reports a year after the interview year. Sixty-three households report an expropriation in the interview year (45 in CHIP 2013 and 18 in CHIP 2018), of whom 59 answer all six items and are in the analysis sample. The six compensation items are cash, pension, medical insurance, Dibao application, housing, and a job arrangement; a household is excluded if any of them is unanswered. No further rule is applied. The rows marked “of which” describe subsets of the analysis sample: the cohort analysis uses households expropriated from 2000 onward, the demographic adjustment of the dispersion analysis requires the head’s age, sex, and education, and the last row is the reference-year sample used as a sensitivity check in Appendix Table A.13.

**Table A.2:** Outcome definitions

| Outcome                                     | Definition                                                                                                                                                                            | Scale       | Role                       |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------|
| Pension                                     | K08_1 = yes                                                                                                                                                                           | binary      | component                  |
| Medical insurance                           | K09_1 = yes                                                                                                                                                                           | binary      | component                  |
| Dibao application                           | applied for urban or rural Dibao at the time of expropriation (K10 in 2013, K11 in 2018)                                                                                              | binary      | component                  |
| Housing                                     | housing item = yes                                                                                                                                                                    | binary      | component                  |
| Employment (job arranged by the government) | replacement-job item = yes                                                                                                                                                            | binary      | component                  |
| Cash                                        | K07_1 = yes                                                                                                                                                                           | binary      | component                  |
| Pension or medical insurance                | Pension OR Medical                                                                                                                                                                    | binary      | supporting evidence        |
| Cash and pension or medical                 | Cash AND (Pension OR Medical)                                                                                                                                                         | binary      | summary-index component    |
| Cash only                                   | Cash AND NOT (Pension OR Medical OR Dibao OR Housing OR Employment)                                                                                                                   | binary      | headline binary outcome    |
| Protection breadth                          | Pension + Medical + Dibao (0-3 count)                                                                                                                                                 | count 0-3   | primary outcome            |
| Any continuing protection                   | 1 if any of Pension, Medical, Dibao, Housing, Employment                                                                                                                              | binary      | headline binary outcome    |
| Summary index                               | equally weighted average of the standardized values of pension or medical insurance, cash and pension or medical, pension, medical insurance, protection breadth, and minus cash only | index units | summary robustness measure |

*Notes.* Each outcome is a formula over the harmonized items in Tables A.3 and A.4. Cash only excludes households that report a pension, medical insurance, Dibao application, housing, or a job arrangement; the 17 households that received cash and whose only protection arrangement is housing (16) or a job arrangement (1) are therefore not coded as cash only. The summary index standardizes each component to mean zero and unit standard deviation in the full analysis sample of 2,239 households before averaging (Kling et al. 2007); the average is not re-standardized, and the same standardization is used in every subsample and stress test.

**Table A.3:** Question wording in the CHIP land-expropriation module

| Item<br>(2013 /<br>2018) | Chinese questionnaire (as<br>administered) | Official English questionnaire                                                                                                                                            | Harmonized<br>variable                | Enters                                                                                     |
|--------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------|
| K05                      | 本户是否有过被征地（征用农民集体所有土地）的经历？                  | Did your household have the experience of land (collective-owned land) expropriation?                                                                                     | sample gate                           | sample definition                                                                          |
| K06_1                    | 被征地的时间                                     | Date of the land expropriation                                                                                                                                            | event year                            | cohort assignment                                                                          |
| K06_2                    | 被征地的总面积                                    | Size of the land expropriation (mu)                                                                                                                                       | land area                             | compensation per mu (Section 6.5)                                                          |
| K07_1                    | 被征地时，是否拿到补偿费（土地补偿费、安置补助费、地上附着物和青苗的补偿费等）？   | Any compensation fees received (land compensation fee, subsidy for resettlement, compensation fee for ground attachments and young crops) due to this land expropriation? | cash                                  | cash only; summary index                                                                   |
| K07_3                    | （接问题7-2）本户实际拿到的征地补偿费数额是                    | (Following K07_2) In which, the actual amount your household received was (yuan)                                                                                          | cash amount received by the household | compensation value (Section 6.5)                                                           |
| K08_1                    | 被征地时，是否办过养老保险？                             | Were there any pension insurance transacted for you or any member of your household due to this land expropriation?                                                       | pension                               | protection breadth; any continuing protection; pension or medical insurance; summary index |
| K08_2                    | 如果办过养老保险的话，其类型是                            | If any kind of pension insurance was transacted, what type is it?                                                                                                         | pension type                          | not an outcome (Section 6.3)                                                               |
| K09_1                    | 被征地时，是否办过医疗保险？                             | Were there any medical insurance transacted for you or any household member due to this land expropriation?                                                               | medical insurance                     | protection breadth; any continuing protection; pension or medical insurance; summary index |
| K10 /<br>K11             | 被征地时，是否申请过低保                               | Did you or any household member apply for any minimum living guarantee (Dibao) due to this land expropriation?                                                            | Dibao application                     | protection breadth; any continuing protection; summary index                               |
| K11 /<br>K12             | 被征地时，政府是否安排住房？                             | Any housing arranged for you or any household member by the government due to this land expropriation?                                                                    | housing                               | any continuing protection (and excluded from cash only)                                    |
| K12 /<br>K13             | 被征地时，政府是否安置就业？                             | Any replacement job arranged for you or any household member due to this land expropriation?                                                                              | job arrangement                       | any continuing protection (and excluded from cash only)                                    |
| - / K10                  | 被征地时，是否办过失业保险？                             | Have you had unemployment insurance due to this land expropriation?                                                                                                       | unemployment insurance (2018 only)    | excluded from every outcome                                                                |

*Notes.* The Chinese wording is transcribed from the rural questionnaires as administered in CHIP 2013 and CHIP 2018; the English wording is transcribed from the official English questionnaires. Where the two waves' items differ only in numbering, the 2013 wording is shown. Every compensation item in Chinese begins with 被征地时 ("at the time of the expropriation"), which the English questionnaire renders as "due to this land expropriation." The Dibao item asks whether the household applied (申请过) for Dibao. Item numbers shift between waves because CHIP 2018 inserts an unemployment-insurance item at K10; that item has no 2013 counterpart and enters no outcome. The "Enters" column refers to sections of this paper.

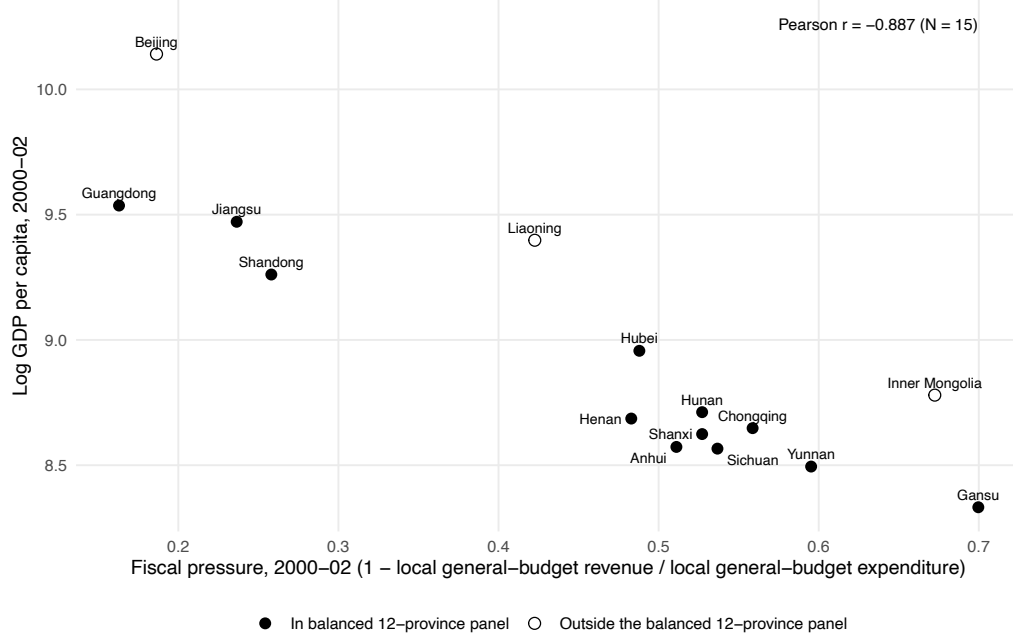
**Table A.4:** Response options and coding rules for the harmonized items

| Item<br>(2013 /<br>2018) | Harmonized<br>variable | Response options, CHIP<br>2013                                                                                                                   | Response options, CHIP<br>2018                                                                                                     | Coding rule                                                                                                                                                                                                                                                    |
|--------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| K05                      | sample gate            | 1 Yes (both arable land and homestead); 2 Yes (arable land only); 3 Yes (homestead only); 4 Yes (land of other usage); 5 No                      | identical to 2013                                                                                                                  | sample gate: keep K05 in {1,2,3,4}; K05 = 5 leaves the module                                                                                                                                                                                                  |
| K06_1                    | event year             | free numeric year                                                                                                                                | free numeric year                                                                                                                  | event_year = K06_1; kept if a calendar year no later than the interview year (2014 for CHIP 2013, 2019 for CHIP 2018); defines the cohort bins                                                                                                                 |
| K06_2                    | land area              | free numeric                                                                                                                                     | free numeric, one decimal                                                                                                          | kept when strictly positive; otherwise missing                                                                                                                                                                                                                 |
| K07_1                    | Cash                   | 1 Yes; 2 No                                                                                                                                      | 1 Yes; 2 No                                                                                                                        | Cash = 1 if K07_1 = 1, 0 if K07_1 = 2, missing otherwise                                                                                                                                                                                                       |
| K07_2,<br>K07_3          | cash amount            | free numeric                                                                                                                                     | free numeric                                                                                                                       | amount = K07_3 if K07_1 = 1 and K07_3 > 0, deflated to 2018 yuan by event-year CPI; K07_2 (total for the land taken) only as a consistency check                                                                                                               |
| K08_1                    | Pension                | 1 Yes; 2 No                                                                                                                                      | 1 Yes; 2 No                                                                                                                        | Pension = 1 if K08_1 = 1, 0 if K08_1 = 2, missing otherwise                                                                                                                                                                                                    |
| K08_2                    | pension type           | 1 pension set up specifically for land-expropriated farmers; 2 urban pension (employee or resident); 3 new rural social insurance; 4 other       | 1 pension set up specifically for land-expropriated farmers; 2 resident endowment insurance; 3 new rural social insurance; 4 other | not used in the outcomes; used in the pension-label stress tests. 2013 code 2 combines urban employee and resident pensions; 2018 code 2 is resident pension only. NRPS is code 3 and the pension set up for land-expropriated farmers is code 1 in both waves |
| K09_1                    | Medical                | 1 Yes; 2 No                                                                                                                                      | 1 Yes; 2 No                                                                                                                        | Medical = 1 if K09_1 = 1, 0 if K09_1 = 2, missing otherwise                                                                                                                                                                                                    |
| K09_2                    | medical type           | 1 medical insurance specifically for land-expropriated farmers; 2 urban medical (employee or resident); 3 new rural cooperative medical; 4 other | 1 medical insurance specifically for land-expropriated farmers; 2 resident; 3 new rural cooperative medical; 4 other               | not used in the outcomes; used in the NCMS stress test (code 3)                                                                                                                                                                                                |
| not asked / K10          | not used               | item does not exist in 2013                                                                                                                      | 1 Yes; 2 No                                                                                                                        | asked only in 2018, so it is excluded from every outcome to keep the two waves comparable                                                                                                                                                                      |
| K10 / K11                | Dibao application      | 1 Yes (urban Dibao); 2 Yes (rural Dibao); 3 No                                                                                                   | identical to 2013                                                                                                                  | Dibao application = 1 if the answer is 1 or 2 (applied for urban or rural Dibao), 0 if 3, missing otherwise                                                                                                                                                    |
| K11 / K12                | Housing                | 1 Yes; 2 No                                                                                                                                      | 1 Yes; 2 No                                                                                                                        | Housing = 1 if the answer is 1, 0 if 2, missing otherwise                                                                                                                                                                                                      |
| K12 / K13                | Employment             | 1 Yes; 2 No                                                                                                                                      | 1 Yes; 2 No                                                                                                                        | Employment = 1 if the answer is 1, 0 if 2, missing otherwise                                                                                                                                                                                                   |

*Notes.* Options are transcribed from the English rural questionnaires; “identical to 2013” means the 2018 option set is unchanged. The rows follow Table A.3. The insurance-type items differ between waves in the same way; for the pension-type item K08\_2, in 2013 code 2 combines urban employee and urban resident pensions, while in 2018 code 2 is the resident pension alone; code 1 (a pension set up for land-expropriated farmers) and code 3 (NRPS) are the same in both waves. Neither type item enters the main outcome definitions. They are used in the measurement stress tests of Section 6.3 (Appendix Table A.13); pension type also identifies the groups in Appendix Tables A.14 and A.18.

## A.2 Fiscal capacity and compensation

Figure A.1 shows the provincial fiscal-pressure measure; Tables A.5–A.7 and Figure A.2 report the cohort-specific coefficients and the timing of their change.



**Figure A.1:** Fiscal pressure and log GDP per capita, 2000–02, for the 15 CHIP provinces. Fiscal pressure is the 2000–02 average of one minus local general-budget revenue divided by local general-budget expenditure; log GDP per capita is the 2000–02 mean of the log of per capita GDP published in the *China Statistical Yearbook*. Filled markers are the 12 provinces of the balanced panel; hollow markers are Beijing, Inner Mongolia, and Liaoning, which have households in every cohort but fewer than five in at least one.

**Table A.5:** Cohort-specific coefficients on 2000–02 fiscal pressure

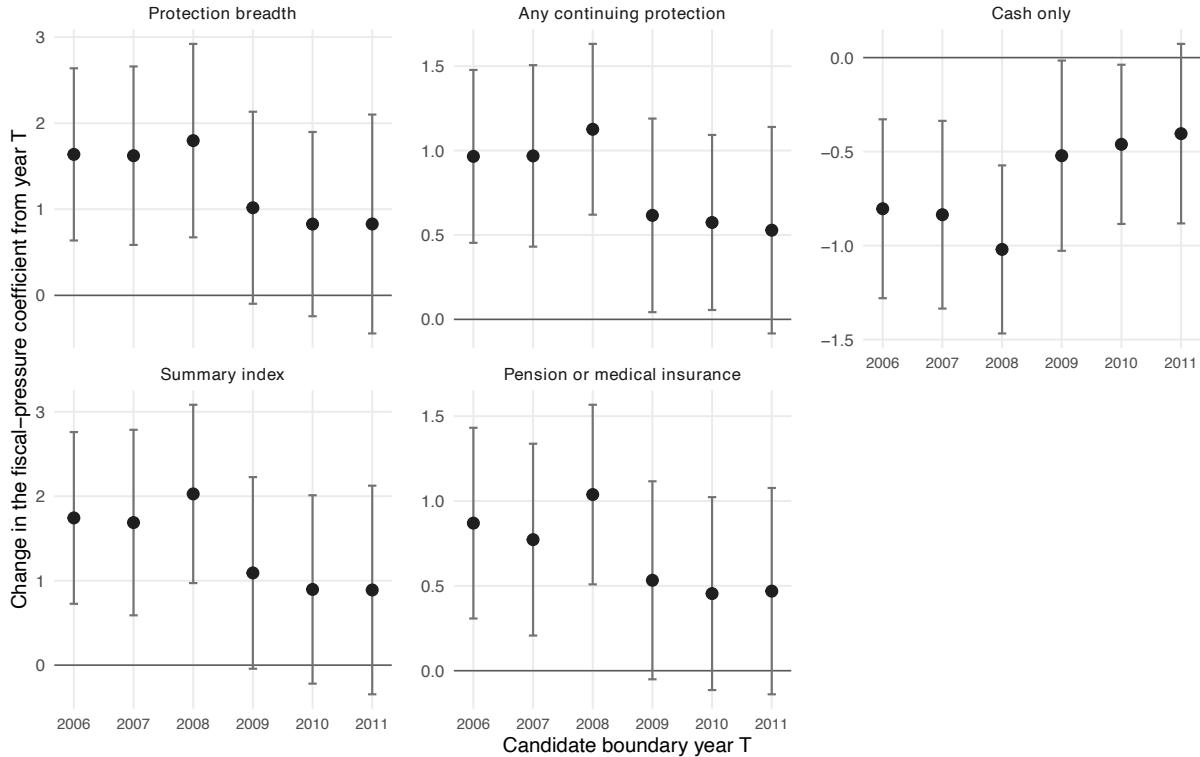
| Cohort                              | Coefficient | (SE)    | 95% CI         | $p$    | $N$ |
|-------------------------------------|-------------|---------|----------------|--------|-----|
| <i>Protection breadth</i>           |             |         |                |        |     |
| 2000–03                             | -2.029**    | (0.737) | [-3.61, -0.45] | 0.016  | 185 |
| 2004–07                             | -1.555**    | (0.548) | [-2.73, -0.38] | 0.013  | 330 |
| 2008–10                             | 0.334       | (0.326) | [-0.37, 1.03]  | 0.323  | 432 |
| 2011–13                             | 0.603       | (0.658) | [-0.81, 2.01]  | 0.375  | 572 |
| 2014+                               | 0.093       | (0.287) | [-0.52, 0.71]  | 0.751  | 603 |
| <i>Any continuing protection</i>    |             |         |                |        |     |
| 2000–03                             | -0.996**    | (0.345) | [-1.74, -0.26] | 0.012  | 185 |
| 2004–07                             | -0.952***   | (0.271) | [-1.53, -0.37] | 0.003  | 330 |
| 2008–10                             | 0.235       | (0.189) | [-0.17, 0.64]  | 0.235  | 432 |
| 2011–13                             | 0.398       | (0.356) | [-0.37, 1.16]  | 0.283  | 572 |
| 2014+                               | 0.047       | (0.288) | [-0.57, 0.67]  | 0.873  | 603 |
| <i>Cash only</i>                    |             |         |                |        |     |
| 2000–03                             | 0.827**     | (0.330) | [0.12, 1.53]   | 0.025  | 185 |
| 2004–07                             | 0.903***    | (0.245) | [0.38, 1.43]   | 0.002  | 330 |
| 2008–10                             | -0.277      | (0.194) | [-0.69, 0.14]  | 0.176  | 432 |
| 2011–13                             | -0.445      | (0.353) | [-1.20, 0.31]  | 0.228  | 572 |
| 2014+                               | 0.083       | (0.236) | [-0.42, 0.59]  | 0.731  | 603 |
| <i>Summary index</i>                |             |         |                |        |     |
| 2000–03                             | -2.078**    | (0.744) | [-3.67, -0.48] | 0.014  | 185 |
| 2004–07                             | -1.930***   | (0.517) | [-3.04, -0.82] | 0.002  | 330 |
| 2008–10                             | 0.280       | (0.332) | [-0.43, 0.99]  | 0.412  | 432 |
| 2011–13                             | 0.554       | (0.689) | [-0.92, 2.03]  | 0.435  | 572 |
| 2014+                               | -0.116      | (0.396) | [-0.96, 0.73]  | 0.773  | 603 |
| <i>Pension or medical insurance</i> |             |         |                |        |     |
| 2000–03                             | -1.068**    | (0.403) | [-1.93, -0.20] | 0.019  | 185 |
| 2004–07                             | -1.059***   | (0.242) | [-1.58, -0.54] | <0.001 | 330 |
| 2008–10                             | 0.063       | (0.178) | [-0.32, 0.44]  | 0.730  | 432 |
| 2011–13                             | 0.229       | (0.351) | [-0.52, 0.98]  | 0.525  | 572 |
| 2014+                               | -0.141      | (0.264) | [-0.71, 0.42]  | 0.601  | 603 |

*Notes.* Estimates of  $\beta_c$  from equation (2), with expropriation-year effects and, where a cohort spans both waves, a survey-wave effect. Standard errors are clustered by province (CR1); confidence intervals and  $p$ -values use the  $t$  distribution with  $G - 1$  degrees of freedom, where  $G$  is the number of provinces in the cohort (15 in every cohort). The 2000–02 fiscal-pressure window overlaps the 2000–03 cohort; for 2004–07 and later cohorts it predates the expropriation. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

**Table A.6:** Change in the fiscal-pressure coefficient at alternative boundary years

| Outcome                      | 2006              | 2007              | 2008              | 2009              | 2010              | 2011              |
|------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Protection breadth           | 1.64<br>[0.0066]  | 1.62<br>[0.0078]  | 1.80<br>[0.0026]  | 1.02<br>[0.0794]  | 0.83<br>[0.1612]  | 0.83<br>[0.2322]  |
| Any continuing protection    | 0.97<br>[0.0107]  | 0.97<br>[0.0076]  | 1.13<br>[0.0011]  | 0.62<br>[0.0392]  | 0.57<br>[0.0095]  | 0.53<br>[0.1071]  |
| Cash only                    | -0.80<br>[0.0161] | -0.84<br>[0.0082] | -1.02<br>[0.0016] | -0.52<br>[0.0417] | -0.46<br>[0.0134] | -0.40<br>[0.1145] |
| Summary index                | 1.74<br>[0.0037]  | 1.69<br>[0.0052]  | 2.03<br>[0.0007]  | 1.09<br>[0.0611]  | 0.90<br>[0.1324]  | 0.89<br>[0.1964]  |
| Pension or medical insurance | 0.87<br>[0.0038]  | 0.77<br>[0.0153]  | 1.04<br>[0.0003]  | 0.53<br>[0.0699]  | 0.45<br>[0.1238]  | 0.47<br>[0.1543]  |

*Notes.* Each entry is the coefficient on fiscal pressure  $\times \mathbf{1}[t \geq T]$  from a regression with province, expropriation-year, and survey-wave effects, estimated on households expropriated from 2000 onward, for boundary year  $T$ . Restricted wild-cluster bootstrap  $p$ -values in brackets, 9,999 draws (the draws used throughout the paper), 15 province clusters. The reference group pools all cohorts before  $T$  and therefore differs from equation (3). The profile is descriptive and is not a formal test for a structural break.

**Figure A.2:** Change in the fiscal-pressure coefficient at alternative boundary years, with 95 percent confidence intervals from province-clustered standard errors and the  $t$  distribution with 14 degrees of freedom. Estimates are those of Table A.6, which also reports the wild-cluster bootstrap  $p$ -values.

**Table A.7:** Fiscal-pressure coefficients with the 2004–07 cohort split

| Outcome            | 2000–03 | 2004–05 | 2006–07 | 2008–10 | 2011–13 | 2014+  |
|--------------------|---------|---------|---------|---------|---------|--------|
| Summary index      | -2.078  | -1.881  | -1.880  | 0.280   | 0.554   | -0.116 |
| Protection breadth | -2.029  | -1.402  | -1.591  | 0.334   | 0.603   | 0.093  |
| Cash only          | 0.827   | 0.868   | 0.904   | -0.277  | -0.445  | 0.083  |
| Pension or medical | -1.068  | -1.096  | -0.981  | 0.063   | 0.229   | -0.141 |

*Notes.* Cohort-specific coefficients  $\beta_c$  from equation (2), with the 2004–07 cohort split into 2004–05 and 2006–07. These are levels of the coefficient, not changes. The two split cohorts contain 151 and 179 households in 14 and 15 provinces.

### A.3 Cross-provincial dispersion

Tables A.8–A.10 and Figure A.3 report the cohort paths of the average provincial gap, the precision checks, and the share of provincial differences accounted for by provincial characteristics. The declines range from 41.3 to 47.2 percent for the random-effects standard deviation and from 19.2 to 24.7 percent under cell-size matching, which cuts the 2004–07 and later cells to the size of the 2000–03 cells and so adds early-period sampling noise to the later cohorts. Resampling households within province–cohort cells gives a 95 percent interval of 20.2 to 47.2 percent for the decline in the average gap in protection breadth, and intervals that exclude zero for all five outcomes; resampling the 12 provinces gives intervals that still exclude zero for any continuing protection and cash only but include zero for protection breadth. Leaving out one province at a time, the smallest decline, 20.3 percent for protection breadth, occurs when Jiangsu is omitted. On all available provinces, which adds Beijing and Liaoning from 2004–07 and Inner Mongolia from 2011–13, the decline is smaller, 20 to 27 percent.

**Table A.8:** Average absolute gap between two provinces, by cohort

| Outcome                      | Average absolute gap between two provinces |         |         |         |       | Decline (%) |      | All provinces   |
|------------------------------|--------------------------------------------|---------|---------|---------|-------|-------------|------|-----------------|
|                              | 2000–03                                    | 2004–07 | 2008–10 | 2011–13 | 2014+ | Average gap | SD   | average gap (%) |
| Protection breadth           | 0.526                                      | 0.482   | 0.311   | 0.340   | 0.278 | 38.5        | 41.8 | 27.3            |
| Any continuing protection    | 0.287                                      | 0.317   | 0.179   | 0.186   | 0.194 | 38.3        | 37.4 | 27.2            |
| Cash only                    | 0.296                                      | 0.270   | 0.189   | 0.181   | 0.160 | 37.6        | 35.1 | 25.9            |
| Summary index                | 0.534                                      | 0.548   | 0.333   | 0.342   | 0.330 | 38.1        | 41.5 | 25.7            |
| Pension or medical insurance | 0.258                                      | 0.320   | 0.199   | 0.190   | 0.199 | 32.1        | 33.5 | 20.0            |

*Notes.* Balanced panel of 12 provinces unless stated. Entries in the first five columns are the average absolute gap between two provinces in demographically adjusted province–cohort means (equation 4). “Decline” is the percentage decline from the average of the two earlier cohorts to the average of the three later cohorts, for the average gap and the standard deviation. The last column repeats the average-gap calculation on all provinces with at least five households in a cohort, which adds Beijing and Liaoning from 2004–07 and Inner Mongolia from 2011–13.

**Table A.9:** Decline in cross-provincial dispersion under alternative precision treatments

| Outcome                      | Average gap | Noise-corrected SD | Random-effects SD | Cell-size matching |
|------------------------------|-------------|--------------------|-------------------|--------------------|
| Protection breadth           | 38.5        | 46.5               | 46.3              | 24.6               |
| Any continuing protection    | 38.3        | 39.1               | 46.1              | 24.7               |
| Cash only                    | 37.6        | 35.7               | 44.6              | 23.3               |
| Summary index                | 38.1        | 45.9               | 47.2              | 24.5               |
| Pension or medical insurance | 32.1        | 35.3               | 41.3              | 19.2               |

*Notes.* Percentage declines from the average of the two earlier cohorts to the average of the three later cohorts, balanced panel of 12 provinces. “Average gap” and “Noise-corrected SD” repeat Table 3. “Random-effects SD” is the standard deviation of the province intercept in a random-intercept model of demographically adjusted household outcomes, estimated by restricted maximum likelihood within each cohort. “Cell-size matching” repeatedly draws, without replacement, as many households in each province-cohort cell from 2004–07 onward as the same province has in the 2000–03 cohort (a cell with fewer households is used whole), recomputes the average gap, and reports the median decline over 2,000 draws; the demographic adjustment is not re-estimated. Because it adds early-period sampling noise to the later cohorts, it is a conservative stress test rather than a corrected estimate.

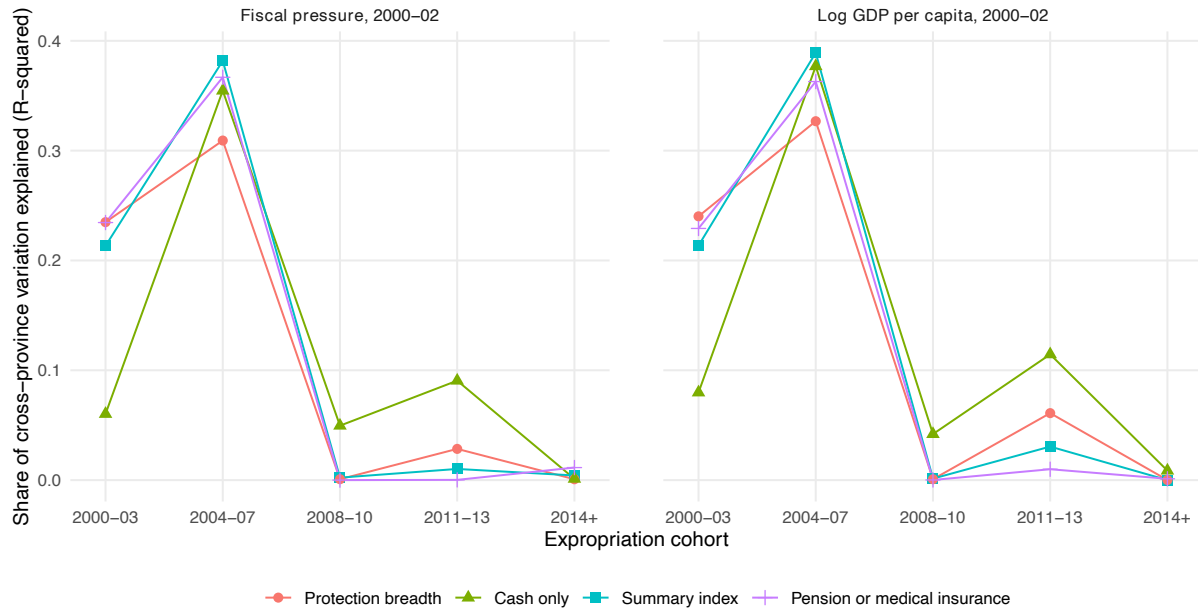
**Table A.10:** Inference and province sensitivity for the decline in the average provincial gap

| <i>Panel A. Bootstrap intervals</i> |                         |                             |                            |  |
|-------------------------------------|-------------------------|-----------------------------|----------------------------|--|
| Outcome                             | Average-gap decline (%) | Household-resampling 95% CI | Province-resampling 95% CI |  |
| Protection breadth                  | 38.5                    | [20.2, 47.2]                | [-23.0, 58.0]              |  |
| Any continuing protection           | 38.3                    | [22.7, 47.5]                | [9.0, 54.5]                |  |
| Cash only                           | 37.6                    | [22.5, 46.9]                | [7.8, 54.2]                |  |
| Summary index                       | 38.1                    | [21.3, 47.2]                | [-14.5, 56.2]              |  |
| Pension or medical insurance        | 32.1                    | [16.8, 43.2]                | [-8.2, 53.0]               |  |

| <i>Panel B. Leaving out one province at a time</i> |          |         |         |         |           |
|----------------------------------------------------|----------|---------|---------|---------|-----------|
| Outcome                                            | Full (%) | Min (%) | Omitted | Max (%) | Omitted   |
| Protection breadth                                 | 38.5     | 20.3    | Jiangsu | 43.6    | Anhui     |
| Any continuing protection                          | 38.3     | 29.0    | Jiangsu | 42.6    | Chongqing |
| Cash only                                          | 37.6     | 28.9    | Jiangsu | 46.1    | Chongqing |
| Summary index                                      | 38.1     | 22.4    | Jiangsu | 43.6    | Anhui     |
| Pension or medical insurance                       | 32.1     | 16.6    | Jiangsu | 39.1    | Anhui     |

*Notes.* Balanced panel of 12 provinces. The decline is the percentage fall in the average absolute gap between two provinces from the two earlier to the three later cohorts. Panel A: household resampling draws households with replacement within each province-cohort cell, holding the 12 provinces fixed; province resampling draws the 12 provinces with replacement, as if they were drawn from a larger population. Both use 2,000 draws and percentile intervals. Demographic adjustment is estimated once on the original sample. Panel B omits one province at a time and recomputes the decline on the remaining 11; it is a sensitivity check, not an inferential procedure.



**Figure A.3:** Share of the cross-provincial variance in adjusted outcomes accounted for by one provincial characteristic at a time, balanced panel of 12 provinces. Each point is the  $R^2$  of a regression of the 12 province-cohort means on 2000-02 fiscal pressure (left) or 2000-02 log GDP per capita (right).

#### A.4 Survey waves and measurement stress tests

Tables A.11–A.14 report the wave-specific estimates, the Dibao cross-check, the measurement stress tests, and the reported pension types.

**Table A.11:** Fiscal-pressure coefficients in the cohorts observed in both survey waves

| <i>Panel A. 2008–10 cohort</i> |                              |                 |                              |                 |
|--------------------------------|------------------------------|-----------------|------------------------------|-----------------|
| Outcome                        | CHIP 2013 $\hat{\beta}$ (SE) | 95% CI          | CHIP 2018 $\hat{\beta}$ (SE) | 95% CI          |
| Protection breadth             | 0.314 (0.532)                | [-0.836, 1.463] | 0.365 (0.374)                | [-0.443, 1.172] |
| Any continuing protection      | 0.192 (0.270)                | [-0.390, 0.775] | 0.310 (0.288)                | [-0.311, 0.932] |
| Cash only                      | -0.262 (0.273)               | [-0.851, 0.327] | -0.302 (0.284)               | [-0.915, 0.312] |
| Summary index                  | 0.251 (0.525)                | [-0.883, 1.386] | 0.317 (0.505)                | [-0.774, 1.408] |
| Pension or medical insurance   | 0.039 (0.257)                | [-0.517, 0.595] | 0.097 (0.299)                | [-0.549, 0.744] |
| <i>Panel B. 2011–13 cohort</i> |                              |                 |                              |                 |
| Outcome                        | CHIP 2013 $\hat{\beta}$ (SE) | 95% CI          | CHIP 2018 $\hat{\beta}$ (SE) | 95% CI          |
| Protection breadth             | 0.670 (0.655)                | [-0.745, 2.085] | 0.547 (0.949)                | [-1.503, 2.597] |
| Any continuing protection      | 0.397 (0.360)                | [-0.381, 1.175] | 0.549 (0.570)                | [-0.682, 1.780] |
| Cash only                      | -0.454 (0.364)               | [-1.241, 0.334] | -0.549 (0.570)               | [-1.780, 0.682] |
| Summary index                  | 0.613 (0.687)                | [-0.872, 2.098] | 0.579 (1.032)                | [-1.649, 2.808] |
| Pension or medical insurance   | 0.252 (0.355)                | [-0.515, 1.019] | 0.271 (0.501)                | [-0.813, 1.354] |
| <i>Panel C. 2014+ cohort</i>   |                              |                 |                              |                 |
| Outcome                        | CHIP 2013 $\hat{\beta}$ (SE) | 95% CI          | CHIP 2018 $\hat{\beta}$ (SE) | 95% CI          |
| Protection breadth             | 0.564 (1.346)                | [-2.399, 3.527] | 0.060 (0.278)                | [-0.536, 0.655] |
| Any continuing protection      | 1.002 (0.635)                | [-0.396, 2.400] | -0.021 (0.288)               | [-0.638, 0.597] |
| Cash only                      | -1.087 (0.526)               | [-2.244, 0.069] | 0.166 (0.237)                | [-0.342, 0.673] |
| Summary index                  | 0.738 (1.284)                | [-2.088, 3.565] | -0.177 (0.383)               | [-0.998, 0.644] |
| Pension or medical insurance   | 0.400 (0.710)                | [-1.162, 1.962] | -0.179 (0.261)               | [-0.740, 0.381] |

*Notes.* Each estimate is equation (2) estimated within one survey wave, with expropriation-year effects and province-clustered standard errors; 95 percent confidence intervals use the  $t$  distribution with  $G - 1$  degrees of freedom. Support is 261 households in CHIP 2013 and 171 in CHIP 2018 for 2008–10, and 448 and 124 for 2011–13, with 14 provinces in every cell (CHIP 2013 has no Inner Mongolia sample and the CHIP 2018 cells have no Beijing households). The sign agrees across waves and the intervals overlap in all ten comparisons of Panels A and B. In Panel C, the CHIP 2013 part of the 2014+ cohort has 42 households in 12 provinces, only three of which have five or more, so its estimates are uninformative; it has no expropriation-year variation and is estimated without year effects.

**Table A.12:** Dibao application at expropriation and Dibao receipt at the survey date

| Wave | Household group                    | Households | Any member on Dibao at the survey date |           |
|------|------------------------------------|------------|----------------------------------------|-----------|
|      |                                    |            | Households                             | Share (%) |
| 2013 | Applied for Dibao at expropriation | 76         | 24                                     | 31.6      |
| 2013 | Did not apply                      | 1,168      | 70                                     | 6.0       |
| 2018 | Applied for Dibao at expropriation | 125        | 25                                     | 20.0      |
| 2018 | Did not apply                      | 870        | 35                                     | 4.0       |

*Notes.* All 2,239 households. Application is the Dibao compensation item (K10 in 2013, K11 in 2018). Current receipt is whether any household member is on Dibao at the survey date, from the household roster. The comparison is descriptive: current receipt can reflect circumstances long after the expropriation.

**Table A.13:** Measurement stress tests for the three main outcomes

| Measurement variant                                                          | <i>N</i> | Early IQR difference |         | $\hat{\theta}$ | Wild <i>p</i> | Average-gap decline (%) |
|------------------------------------------------------------------------------|----------|----------------------|---------|----------------|---------------|-------------------------|
|                                                                              |          | 2000–03              | 2004–07 |                |               |                         |
| <i>Protection breadth</i>                                                    |          |                      |         |                |               |                         |
| Main estimates                                                               | 2,239    | -0.42                | -0.32   | 1.625          | 0.0053        | 38.5                    |
| Exclude households with an NRPS label for an expropriation in or before 2008 | 2,170    | -0.35                | -0.23   | 1.031          | 0.2435        | 30.9                    |
| Do not count NRPS-labeled pensions                                           | 2,239    | -0.36                | -0.25   | 1.373          | 0.0117        | 47.7                    |
| Both of the above                                                            | 2,170    | -0.35                | -0.23   | 1.213          | 0.0913        | 47.2                    |
| Exclude Dibao                                                                | 2,239    | -0.45                | -0.35   | 1.425          | 0.0276        | 38.8                    |
| Exclude Dibao and the NRPS-label households                                  | 2,170    | -0.37                | -0.25   | 0.807          | 0.4735        | 30.3                    |
| Exclude Dibao; do not count NRPS-labeled pensions                            | 2,239    | -0.38                | -0.27   | 1.173          | 0.0861        | 50.3                    |
| Do not count NCMS-labeled medical insurance                                  | 2,239    | -0.34                | -0.14   | 1.241          | 0.0079        | 42.7                    |
| Expropriations up to the reference year only (2013 / 2018)                   | 2,180    | -0.42                | -0.32   | 1.647          | 0.0077        | 38.2                    |
| <i>Any continuing protection</i>                                             |          |                      |         |                |               |                         |
| Main estimates                                                               | 2,239    | -20.7                | -19.7   | 0.937          | 0.0252        | 38.3                    |
| Exclude households with an NRPS label for an expropriation in or before 2008 | 2,170    | -17.4                | -16.4   | 0.664          | 0.0768        | 32.5                    |
| Do not count NRPS-labeled pensions                                           | 2,239    | -20.7                | -20.3   | 1.061          | 0.0060        | 40.0                    |
| Both of the above                                                            | 2,170    | -17.4                | -16.4   | 0.806          | 0.0129        | 35.7                    |
| Exclude Dibao                                                                | 2,239    | -22.3                | -22.1   | 0.754          | 0.0492        | 35.4                    |
| Exclude Dibao and the NRPS-label households                                  | 2,170    | -18.6                | -18.9   | 0.456          | 0.4096        | 27.8                    |
| Exclude Dibao; do not count NRPS-labeled pensions                            | 2,239    | -22.3                | -22.6   | 0.883          | 0.0119        | 37.4                    |
| Do not count NCMS-labeled medical insurance                                  | 2,239    | -20.4                | -11.2   | 0.974          | 0.0166        | 34.9                    |
| Expropriations up to the reference year only (2013 / 2018)                   | 2,180    | -20.7                | -19.7   | 0.933          | 0.0214        | 37.3                    |
| <i>Cash only</i>                                                             |          |                      |         |                |               |                         |
| Main estimates                                                               | 2,239    | 17.1                 | 18.7    | -0.847         | 0.0225        | 37.6                    |
| Exclude households with an NRPS label for an expropriation in or before 2008 | 2,170    | 14.0                 | 16.2    | -0.591         | 0.0465        | 32.3                    |
| Do not count NRPS-labeled pensions                                           | 2,239    | 17.1                 | 19.2    | -0.972         | 0.0071        | 38.5                    |
| Both of the above                                                            | 2,170    | 14.0                 | 16.2    | -0.733         | 0.0138        | 34.4                    |
| Exclude Dibao                                                                | 2,239    | 17.8                 | 21.1    | -0.672         | 0.0593        | 31.5                    |
| Exclude Dibao and the NRPS-label households                                  | 2,170    | 14.2                 | 18.6    | -0.391         | 0.2926        | 26.1                    |
| Exclude Dibao; do not count NRPS-labeled pensions                            | 2,239    | 17.8                 | 21.6    | -0.798         | 0.0333        | 32.8                    |
| Do not count NCMS-labeled medical insurance                                  | 2,239    | 16.9                 | 9.7     | -0.828         | 0.0201        | 37.3                    |
| Expropriations up to the reference year only (2013 / 2018)                   | 2,180    | 17.1                 | 18.7    | -0.822         | 0.0188        | 37.6                    |

*Notes.* “Early IQR difference” is the cohort-specific coefficient from equation (2) multiplied by 0.207, in protection items for breadth and percentage points otherwise.  $\hat{\theta}$  and its wild-cluster *p*-value are from equation (3). “Average-gap decline” is the percentage decline in the average absolute gap between two provinces on the balanced 12-province panel. Every row uses the same 9,999 bootstrap draws as Table 2, to reduce Monte Carlo noise in comparisons across specifications. The demographic adjustment is re-estimated for each row, and the 12 provinces of the main panel are held fixed; every cell of every row has at least five households. “Exclude households with an NRPS label for an expropriation in or before 2008” drops the 69 households with pre-rollout NRPS-labeled pension reports. “Do not count NRPS-labeled pensions” sets the pension indicator to zero for every NRPS-labeled pension and recomputes all outcomes. “Exclude Dibao” removes Dibao application from every outcome. “Do not count NCMS-labeled medical insurance” sets the medical indicator to zero for the 540 medical reports whose type is the New Rural Cooperative Medical Scheme and recomputes all outcomes. “Expropriations up to the reference year only” keeps the 2,180 households whose expropriation year is no later than 2013 (CHIP 2013) or 2018 (CHIP 2018). Like the other rows, these are stress tests, not preferred codings.

**Table A.14:** Reported pension type among households reporting a pension

| <i>Panel A. By expropriation cohort</i> |            |                 |                                                   |          |                                 |       |                               |                 |
|-----------------------------------------|------------|-----------------|---------------------------------------------------|----------|---------------------------------|-------|-------------------------------|-----------------|
| Cohort                                  | Households | Pension reports | Reported pension type, $N$ (% of pension reports) |          |                                 |       | Timing of the label           |                 |
|                                         |            |                 | Dedicated scheme                                  | NRPS     | Urban or resident (2013 / 2018) | Other | NRPS label, event $\leq$ 2008 | Other ambiguous |
| 1995–99                                 | 117        | 26              | 7 (27)                                            | 13 (50)  | 6 (23)                          | 0     | 13 (50)                       | 2 (8)           |
| 2000–03                                 | 185        | 46              | 13 (28)                                           | 10 (22)  | 22 (48)                         | 1     | 10 (22)                       | 0 (0)           |
| 2004–07                                 | 330        | 67              | 17 (25)                                           | 26 (39)  | 23 (34)                         | 1     | 26 (39)                       | 5 (7)           |
| 2008–10                                 | 432        | 118             | 28 (24)                                           | 61 (52)  | 25 (21)                         | 4     | 20 (17)                       | 31 (26)         |
| 2011–13                                 | 572        | 237             | 51 (22)                                           | 155 (65) | 27 (11)                         | 4     | 0 (0)                         | 3 (1)           |
| 2014+                                   | 603        | 212             | 51 (24)                                           | 145 (68) | 11 (5)                          | 5     | 0 (0)                         | 0 (0)           |
| All                                     | 2,239      | 706             | 167 (24)                                          | 410 (58) | 114 (16)                        | 15    | 69 (10)                       | 41 (6)          |

| <i>Panel B. Pre-2009 NRPS labels by province</i> |                         |                        |                               |                        |                   |  |
|--------------------------------------------------|-------------------------|------------------------|-------------------------------|------------------------|-------------------|--|
| Province                                         | Fiscal pressure 2000–02 | Pension reports        |                               |                        | NRPS share,       |  |
|                                                  |                         | All, event $\leq$ 2007 | NRPS label, event $\leq$ 2007 | NRPS label, event 2008 | event $\leq$ 2007 |  |
| Guangdong                                        | 0.163                   | 10                     | 9                             | 0                      | 0.90              |  |
| Beijing                                          | 0.186                   | 1                      | 1                             | 1                      | 1.00              |  |
| Jiangsu                                          | 0.236                   | 80                     | 28                            | 1                      | 0.35              |  |
| Shandong                                         | 0.258                   | 12                     | 4                             | 3                      | 0.33              |  |
| Liaoning                                         | 0.423                   | 0                      | 0                             | 0                      | –                 |  |
| Henan                                            | 0.483                   | 0                      | 0                             | 0                      | –                 |  |
| Hubei                                            | 0.488                   | 1                      | 1                             | 0                      | 1.00              |  |
| Anhui                                            | 0.511                   | 5                      | 0                             | 0                      | 0.00              |  |
| Hunan                                            | 0.527                   | 0                      | 0                             | 0                      | –                 |  |
| Shanxi                                           | 0.527                   | 2                      | 1                             | 7                      | 0.50              |  |
| Sichuan                                          | 0.537                   | 14                     | 0                             | 0                      | 0.00              |  |
| Chongqing                                        | 0.559                   | 8                      | 1                             | 0                      | 0.12              |  |
| Yunnan                                           | 0.595                   | 5                      | 3                             | 0                      | 0.60              |  |
| Inner Mongolia                                   | 0.672                   | 0                      | 0                             | 0                      | –                 |  |
| Gansu                                            | 0.700                   | 1                      | 1                             | 8                      | 1.00              |  |

*Notes.* Pension type is the answer to K08\_2 among households answering yes to K08\_1. “Dedicated scheme” is a pension set up for land-expropriated farmers (code 1); “Urban or resident” is 2013 code 2 (urban employee or resident pension) or 2018 code 2 (resident pension). “NRPS label, event  $\leq$  2008” counts NRPS reports for expropriations before the national NRPS guideline of September 2009 (the pre-rollout reports discussed in Section 6.3). “Other ambiguous” counts NRPS reports for 2009 expropriations, whose month is unknown, and 2018 resident-pension reports for expropriations in or before 2013, before the national merger of the resident schemes in 2014; some localities had merged schemes earlier. The 2009 NRPS reports enter the stress test that sets every NRPS-labeled pension to zero; the 2018 resident-pension reports enter no stress test. In Panel B, the last column is the share of pension reports for expropriations in or before 2007 that carry an NRPS label.

## A.5 Household characteristics, recall, and compensation value

Tables A.15–A.18 and Figure A.4 report the reweighting, recall-period, and compensation-value checks and the illustrative pension calibration.

**Table A.15:** Decline in dispersion after reweighting later cohorts to early-cohort composition

| <i>Panel A. Decline in dispersion</i> |                         |            |                |            |
|---------------------------------------|-------------------------|------------|----------------|------------|
| Outcome                               | Average-gap decline (%) |            | SD decline (%) |            |
|                                       | Unweighted              | Reweighted | Unweighted     | Reweighted |
| Summary index                         | 38.1                    | 33.5       | 41.5           | 37.4       |
| Protection breadth                    | 38.5                    | 33.9       | 41.8           | 38.3       |
| Cash only                             | 37.6                    | 26.7       | 35.1           | 24.8       |
| Pension or medical                    | 32.1                    | 26.9       | 33.5           | 27.9       |

| <i>Panel B. Characteristics of the head (education shares in percent)</i> |               |                   |                   |
|---------------------------------------------------------------------------|---------------|-------------------|-------------------|
| Head's education                                                          | Early cohorts | Later, unweighted | Later, reweighted |
| No schooling                                                              | 5.6           | 5.0               | 6.4               |
| Primary                                                                   | 32.3          | 29.1              | 32.7              |
| Junior high                                                               | 47.3          | 51.2              | 47.2              |
| Senior high                                                               | 10.5          | 10.3              | 9.6               |
| Vocational high / technical                                               | 1.2           | 0.4               | 1.0               |
| Secondary specialized                                                     | 1.4           | 1.5               | 1.4               |
| Junior college                                                            | 1.2           | 1.8               | 1.1               |
| University                                                                | 0.6           | 0.6               | 0.6               |
| Graduate                                                                  | 0.0           | 0.1               | 0.0               |
| Age at expropriation, standardized difference                             |               | -0.66             | +0.01             |
| Female head, standardized difference                                      |               | +0.05             | +0.02             |

*Notes.* Later cohorts (2008 onward) are reweighted to the 514 households expropriated in 2000–07 by inverse-probability weights from a logit of early-cohort membership on the head's sex, education (category indicators), age at expropriation, and its square (DiNardo et al. 1996); the demographic adjustment and the provincial means are then computed with these weights. The standardized mean difference in age at expropriation falls from  $-0.66$  to  $+0.01$ ; the effective sample is 881 of 1,606 later households and the largest weight is 2.88. No trimming. Panel B compares early households with later households before and after reweighting.

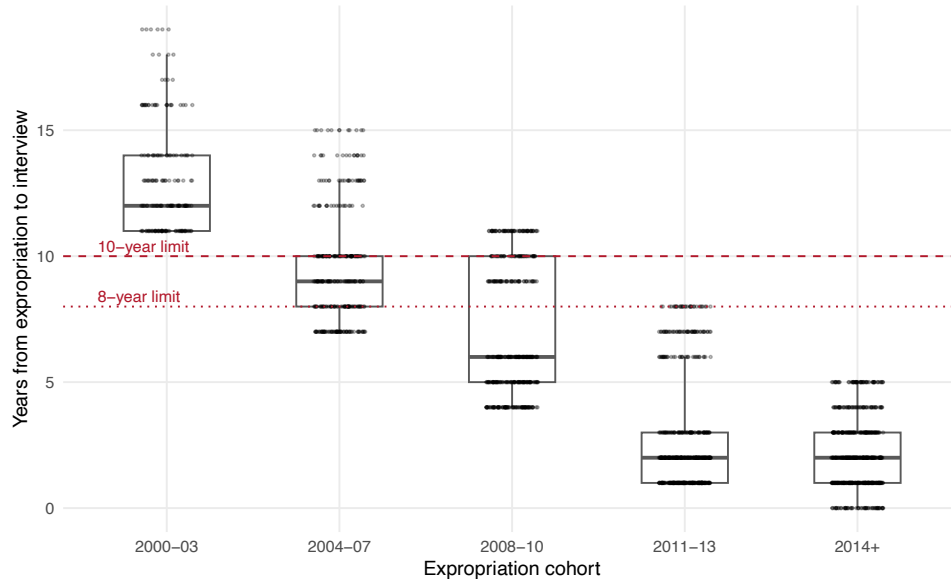
**Table A.16:** Recall-period restrictions, with 2004–07 as the early window

| <i>Panel A. Decline in the average provincial gap (%)</i> |              |                 |              |                |
|-----------------------------------------------------------|--------------|-----------------|--------------|----------------|
| Outcome                                                   | 13 provinces |                 | 8 provinces  |                |
|                                                           | Unrestricted | $\leq 10$ years | Unrestricted | $\leq 8$ years |
| Summary index                                             | 27.8         | 35.3            | 12.9         | 33.9           |
| Protection breadth                                        | 23.6         | 30.0            | 9.0          | 31.9           |
| Cash only                                                 | 22.4         | 33.6            | 1.2          | 27.8           |
| Pension or medical                                        | 29.2         | 36.4            | 5.9          | 29.0           |

| <i>Panel B. Fiscal-pressure coefficient for the 2004–07 cohort under the same restrictions</i> |                                 |                                    |                                   |
|------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------|-----------------------------------|
| Outcome                                                                                        | Unrestricted $\hat{\beta}$ (SE) | $\leq 10$ years $\hat{\beta}$ (SE) | $\leq 8$ years $\hat{\beta}$ (SE) |
| Protection breadth                                                                             | -1.555 (0.548)                  | -2.090 (0.629)                     | -2.089 (0.690)                    |
| Any continuing protection                                                                      | -0.952 (0.271)                  | -1.346 (0.310)                     | -1.177 (0.287)                    |
| Cash only                                                                                      | 0.903 (0.245)                   | 1.196 (0.281)                      | 1.157 (0.269)                     |
| Summary index                                                                                  | -1.930 (0.517)                  | -2.586 (0.590)                     | -2.494 (0.561)                    |
| Pension or medical insurance                                                                   | -1.059 (0.242)                  | -1.443 (0.270)                     | -1.332 (0.233)                    |

*Notes.* Recall is the number of calendar years between the expropriation and the interview (2014 for CHIP 2013, 2019 for CHIP 2018). The early window is 2004–07 in every column, because no 2000–03 expropriation lies within ten years of the interview. In Panel A, each restriction is compared with the unrestricted sample on the same provinces: those with at least five households in 2004–07 and in each later cohort under the restriction (13 provinces for ten years, 8 for eight years); every cell has at least five households, and at least seven in the ten-year comparison. A five-year limit would remove every cohort before 2008 and is not reported. Panel B estimates equation (2) on the 2004–07 cohort alone with province-clustered standard errors. Support falls from 330 households in 15 provinces to 257 in 14 provinces and 133 in 13 provinces; both restricted columns contain only CHIP 2013 households, because a 2004–07 expropriation observed in 2018 is at least twelve years old.

**Figure A.4:** Years between expropriation and interview, by expropriation cohort, with the ten-year and eight-year limits used in Table A.16. Interviews took place in 2014 (CHIP 2013) and 2019 (CHIP 2018).

**Table A.17:** Real compensation and provincial fiscal pressure

| <i>Panel A. Reported amounts by cohort (2018 yuan)</i> |          |                         |                          |               |  |
|--------------------------------------------------------|----------|-------------------------|--------------------------|---------------|--|
| Cohort                                                 | <i>N</i> | Reporting an amount (%) | Median real compensation | Median per mu |  |
| 2000–03                                                | 185      | 84.9                    | 15,013                   | 11,913        |  |
| 2004–07                                                | 330      | 90.3                    | 20,720                   | 15,731        |  |
| 2008–10                                                | 432      | 94.2                    | 23,656                   | 13,696        |  |
| 2011–13                                                | 572      | 90.9                    | 27,489                   | 19,626        |  |
| 2014+                                                  | 603      | 84.4                    | 15,311                   | 13,481        |  |

| <i>Panel B. Coefficient on fiscal pressure</i> |                          |        |                          |        |                |               |
|------------------------------------------------|--------------------------|--------|--------------------------|--------|----------------|---------------|
| Outcome                                        | Expropriated before 2008 |        | Expropriated 2008 onward |        | Pooled change  |               |
|                                                | $\hat{\beta}$ (SE)       | IQR    | $\hat{\beta}$ (SE)       | IQR    | $\hat{\theta}$ | Wild <i>p</i> |
| Log real compensation                          | -1.782 (0.422)           | -0.369 | 1.397 (0.905)            | +0.290 | 3.332          | 0.2650        |
| Log real compensation per mu                   | -0.524 (0.613)           | -0.109 | 2.129 (1.124)            | +0.442 | 2.125          | 0.5596        |

*Notes.* Compensation is the amount the household reports actually receiving (K07\_3), counted when the household reports receiving cash (K07\_1) and an amount above zero, and deflated to 2018 yuan with the consumer price index of the expropriation year (World Bank 2026); per mu divides it by the area taken when the area is positive. Amounts are not trimmed. Excluding the 31 households whose reported amount exceeds the total compensation for the land (K07\_2) leaves the early IQR difference in log compensation at  $-0.37$ . Panel B estimates equation (2) on the two earlier cohorts together and on the three later cohorts together, with expropriation-year and survey-wave effects and province-clustered standard errors. “IQR” multiplies the coefficient by 0.207, in log points; a negative entry means that households in fiscally weaker provinces reported less cash.  $\hat{\theta}$  is the pooled change from equation (3), with a restricted wild-cluster bootstrap *p*-value, 9,999 draws. Every regression has 15 province clusters. The regressions use 455 (before 2008) and 1,436 (2008 onward) households for log compensation and 400 and 1,286 for log compensation per mu; the pooled changes use 1,979 and 1,766 households, including those expropriated in 1995–99.

**Table A.18:** Illustrative discounted value of benchmark pension flows

| Pension type                              | CHARLS<br>annual median | Discounted value, delay 10, duration 15 |         |         | Range at 2%,<br>delay 5–15,<br>duration 10–20 | CHIP median<br>cash ( <i>N</i> ) | Ratio at<br>2% |
|-------------------------------------------|-------------------------|-----------------------------------------|---------|---------|-----------------------------------------------|----------------------------------|----------------|
|                                           |                         | 2%<br>(preferred)                       | 3%      | 5%      |                                               |                                  |                |
| New Rural Pension Scheme                  | 648                     | 6,828                                   | 5,754   | 4,127   | 4,323–9,593                                   | 20,301 (362)                     | 0.34           |
| Pension for land-expropriated farmers     | 5,398                   | 56,897                                  | 47,948  | 34,395  | 36,025–79,940                                 | 24,679 (147)                     | 2.31           |
| <i>Other pension types, for reference</i> |                         |                                         |         |         |                                               |                                  |                |
| Old rural pension (pre-2009 scheme)       | 594                     | 6,259                                   | 5,274   | 3,783   | 3,963–8,793                                   | –                                | –              |
| Urban-rural resident pension              | 702                     | 7,397                                   | 6,233   | 4,471   | 4,683–10,392                                  | –                                | –              |
| Urban resident pension                    | 9,662                   | 101,845                                 | 85,826  | 61,568  | 64,486–143,093                                | –                                | –              |
| Employee basic pension                    | 18,352                  | 193,449                                 | 163,022 | 116,944 | 122,486–271,797                               | –                                | –              |
| Other pension                             | 3,401                   | 35,845                                  | 30,207  | 21,669  | 22,696–50,362                                 | –                                | –              |

*Notes.* All values are in 2010 yuan. (1) Annual pension flows are external benchmarks: 12 times the median current monthly amount among CHARLS 2013 respondents receiving each pension (Zhao et al. 2014), deflated from 2013 to 2010 prices with the same consumer price index used for CHIP cash. NRPS uses respondents receiving NRPS benefits (items FN075 and FN077); the pension for land-expropriated farmers uses respondents receiving it (FN079); the old rural pension, the urban–rural resident pension, and the urban resident pension use the corresponding answers to FN057–FN068; the employee pension uses FN002–FN005; and other pensions use FN083 and FN096. Medians are unweighted, and a respondent receiving two types counts in both. (2) CHIP records the pension type but not the household’s benefit level, the number of beneficiaries, or their ages. (3) The discounted value is  $\sum_{t=d+1}^{d+T} B/(1+r)^t$  for annual benefit  $B$ , real discount rate  $r$ , a wait of  $d$  years after the expropriation, with the first payment at the end of year  $d+1$ , and duration  $T$ ; the preferred values are  $r = 2\%$ ,  $d = 10$ ,  $T = 15$ , and the range column varies  $d$  over 5, 10, and 15 years and  $T$  over 10, 15, and 20 years at  $r = 2\%$ . It is the discounted value of a benchmark pension flow for one beneficiary under these assumptions, not matched household pension wealth, net benefit value, or government cost. (4) It includes no mortality, survivor benefits, indexation, contributions, or the insurance value of an annuity. (5) It is not a welfare ranking of cash and pensions. (6) The CHIP comparator is the median cash compensation actually received (K07\_3), deflated with the CPI of the expropriation year, among analysis households reporting that pension type and a positive amount; the number of such households is in parentheses. Across all 2,239 analysis households the median is 17,000 yuan (1,979 households reporting an amount), and among households reporting any pension it is 20,814 yuan (624). The ratio divides the preferred discounted value by that median. The full grid of 27 assumption combinations is in the replication files.

## A.6 Alternative definitions and remaining checks

Tables A.19–A.22 report the checks on land type, housing, alternative definitions, and trade exposure.

**Table A.19:** Type of land expropriated and the arable-land subsample

| <i>Panel A. Type of land by cohort (percent of households)</i> |          |                      |             |                |             |  |
|----------------------------------------------------------------|----------|----------------------|-------------|----------------|-------------|--|
| Cohort                                                         | <i>N</i> | Arable and homestead | Arable only | Homestead only | Other usage |  |
| Full sample                                                    | 2,239    | 15.5                 | 78.1        | 1.2            | 5.1         |  |
| 2000–03                                                        | 185      | 10.8                 | 84.3        | 0.5            | 4.3         |  |
| 2004–07                                                        | 330      | 19.4                 | 76.4        | 1.2            | 3.0         |  |
| 2008–10                                                        | 432      | 15.5                 | 77.5        | 0.9            | 6.0         |  |
| 2011–13                                                        | 572      | 15.2                 | 78.7        | 1.0            | 5.1         |  |
| 2014+                                                          | 603      | 13.8                 | 77.4        | 1.8            | 7.0         |  |

| <i>Panel B. Main results re-estimated on households whose expropriation involved arable land</i> |                         |        |                                                      |        |                        |  |
|--------------------------------------------------------------------------------------------------|-------------------------|--------|------------------------------------------------------|--------|------------------------|--|
| Outcome                                                                                          | Average-gap decline (%) |        | Change in fiscal-pressure coefficient $\hat{\theta}$ |        |                        |  |
|                                                                                                  | Full                    | Arable | Full                                                 | Arable | Wild <i>p</i> , arable |  |
| Protection breadth                                                                               | 38.5                    | 38.6   | 1.625                                                | 1.900  | 0.0030                 |  |
| Any continuing protection                                                                        | 38.3                    | 35.6   | 0.937                                                | 1.117  | 0.0135                 |  |
| Cash only                                                                                        | 37.6                    | 29.9   | -0.847                                               | -1.018 | 0.0138                 |  |
| Summary index                                                                                    | 38.1                    | 37.7   | 1.472                                                | 1.727  | 0.0086                 |  |
| Pension or medical insurance                                                                     | 32.1                    | 35.0   | 0.653                                                | 0.767  | 0.0723                 |  |

*Notes.* Panel A covers all 2,239 households; the full-sample row includes the 117 households expropriated in 1995–99. Panel B compares the full sample with the 2,097 households whose expropriation involved arable land (the first two categories of Panel A). The decline is the percentage fall in the average absolute gap between two provinces on the 12 provinces of the main panel, each of which has at least five arable-land households in every cohort.  $\hat{\theta}$  is from equation (3); wild-cluster *p*-values use 9,999 restricted bootstrap draws over 15 province clusters.

**Table A.20:** Housing compensation

*Panel A. Housing arranged by the government, by cohort*

| Cohort  | <i>N</i> | Housing receipt (%) |
|---------|----------|---------------------|
| 2000–03 | 185      | 2.7                 |
| 2004–07 | 330      | 7.6                 |
| 2008–10 | 432      | 3.7                 |
| 2011–13 | 572      | 4.7                 |
| 2014+   | 603      | 2.0                 |

*Panel B. Decline in the average provincial gap with and without housing*

| Measure                                      | Average-gap decline (%) |
|----------------------------------------------|-------------------------|
| Any continuing protection, main definition   | 38.3                    |
| Any continuing protection, excluding housing | 33.5                    |
| Protection breadth (never includes housing)  | 38.5                    |

*Notes.* Panel A covers the 2,122 households expropriated from 2000 onward. Panel B recomputes the decline on the balanced 12-province panel with housing removed from any continuing protection; protection breadth never includes housing. Seventeen of the 2,122 households report housing as their only protection arrangement.

**Table A.21:** Decline in the average provincial gap for alternative outcome definitions

| Class  | Outcome                     | Average gap, pre-2008 | Average gap, 2008 onward | Decline (%) |
|--------|-----------------------------|-----------------------|--------------------------|-------------|
| Coarse | Cash only                   | 0.283                 | 0.177                    | 37.6        |
| Coarse | Any continuing protection   | 0.302                 | 0.186                    | 38.3        |
| Fine   | Medical                     | 0.281                 | 0.160                    | 42.9        |
| Fine   | Cash and pension or medical | 0.263                 | 0.161                    | 38.7        |
| Fine   | Protection breadth          | 0.504                 | 0.310                    | 38.5        |
| Fine   | Summary index               | 0.541                 | 0.335                    | 38.1        |
| Fine   | Pension or medical          | 0.289                 | 0.196                    | 32.1        |
| Fine   | Pension                     | 0.248                 | 0.186                    | 24.8        |

*Notes.* Balanced panel of 12 provinces. “Average gap, pre-2008” averages the 2000–03 and 2004–07 cohorts; “Average gap, 2008 onward” averages the three later cohorts. Coarse measures require only the distinction between cash only and any continuing protection; fine measures use individual components or their combinations.

**Table A.22:** Trade exposure before the 2008 crisis and the change in the fiscal-pressure coefficient

| Outcome            | Change from 2008 onward, $\hat{\theta}$ |               | Magnitude<br>reduced (%) | Wild-cluster <i>p</i> |               |
|--------------------|-----------------------------------------|---------------|--------------------------|-----------------------|---------------|
|                    | Main                                    | With exposure |                          | Main                  | With exposure |
| Summary index      | 1.941                                   | 0.881         | 55                       | 0.0023                | 0.0676        |
| Protection breadth | 1.977                                   | 1.164         | 41                       | 0.0021                | 0.0235        |
| Cash only          | -0.841                                  | -0.285        | 66                       | 0.0480                | 0.4098        |
| Pension or medical | 0.918                                   | 0.259         | 72                       | 0.0035                | 0.2479        |

*Notes.* “Main” is  $\hat{\theta}$  from equation (3) estimated on the 2,122 households expropriated from 2000 onward; it therefore differs from Table 2, whose reference group also includes the 117 households expropriated in 1995–99. “With exposure” adds trade openness  $\times$  expropriated from 2008 onward to the same equation and sample. Trade openness is the 2005–07 average of provincial goods imports plus exports, by location of the operating unit, converted from US dollars at the annual-average exchange rate and divided by GDP, from the *China Statistical Yearbook* (National Bureau of Statistics of China 2001–2008). Across the 15 provinces it correlates with 2000–02 fiscal pressure at  $-0.854$  and with 2000–02 log GDP per capita at  $+0.872$ . Wild-cluster *p*-values test  $\hat{\theta}$  with 9,999 restricted bootstrap draws; in the exposure column the trade interaction is kept in both the restricted and the unrestricted model.

## Data availability

The CHIP 2013 and 2018 household microdata are distributed by the China Institute for Income Distribution at Beijing Normal University to registered users under its data-use agreement, and the CHARLS 2013 data (version 20151118) are distributed by the CHARLS project at Peking University to registered users. The author is not permitted to redistribute either. The provincial fiscal, GDP and trade series are from the *China Statistical Yearbook* published by the National Bureau of Statistics of China, and the consumer price index is from the World Bank’s World Development Indicators. The code that constructs every table and figure from these sources, the derived provincial variables with their source references, and the documentation of the questionnaire mapping will be provided in a replication package in accordance with the journal’s data and code policy; the restricted microdata are not included.

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## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used ChatGPT (OpenAI) and Claude (Anthropic) in order to edit the language of the manuscript and to draft and debug the analysis code. After using these tools, the author reviewed and edited the content and the code as needed and takes full responsibility for the analysis and for the content of the published article.

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